

JioMausam

Complete Q&A prep

Everything behind our video and our presentation, explained simply: every term, every number, how we tested the links, what changed and why, every cross-question we can think of, and the backup plan for each assumption.

Read Parts 1–4 for the story, Parts 5–6 for the evidence, Part 7 whenever a word is unclear. Drill Part 8 out loud: answer, then stop. Know Part 9 for any “what if it fails?”. Keep Part 10 open the night before.

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JioMausam explained like you're 10

No jargon. If you can explain this page to a 10-year-old, you can explain it to a judge.

1 • The problem

Imagine one city where it is pouring in one neighbourhood and dry twenty kilometres away. That really happened in Mumbai on 26 July 2005: Santacruz got 944 mm of rain in a day, Colaba got 73. Now imagine you run a delivery company. Your phone says "Heavy rain in Mumbai". Do you stop every rider in the city? Or keep everyone going? You don't know **which streets** are flooded. That is the problem: there is weather information, but it is too blurry to make decisions street by street.

2 • The surprising idea

Mobile towers talk to each other with invisible radio beams called **microwave links**, like a torch shining from one tower to another. When it rains, the raindrops soak up some of the beam, so less of it arrives at the other end. Think of shouting across a playground while it's noisy: the louder the noise, the less of your voice gets across. If you measure how much weaker the beam got, you can estimate how hard it rained along that line. Jio has lots of towers, so Jio **may** already own a giant rain-measuring machine without knowing it.

3 • But only some towers can do it

Not every tower uses these beams. Many are connected by **fibre** cables in the ground, and cables don't feel rain. So the first question is simply: how many of Jio's towers use beams, and where? Nobody outside Jio knows yet.

4 • Did we check if it works?

Yes. Scientists proved the idea in 2006 (Israel) and made rain maps of a whole country in 2013 (the Netherlands). We wanted to check it ourselves, so we took a free research dataset from a university in Germany: 500 real tower beams, measured every minute for 11 days, plus weather radar as the "answer key". We turned the beam signals into rain and compared. The beams followed the radar closely. But when we asked a strict question ("did it rain at least 10 mm in 3 hours in this neighbourhood today?"), the beams alone caught only **68 out of every 100** real heavy-rain events. Our target is 80. So beams alone are not good enough.

5 • So we team them up

If one friend is sometimes wrong, you ask several friends. JioMausam combines the tower beams with weather radar (IMD), rain buckets called gauges (IMD and the Mumbai city corporation, BMC), satellites as a backup, and Jio's own records of broken equipment (so a broken tower isn't mistaken for rain). Then we test every combination and keep the cheapest one that reaches our target.

6 • Who uses it first?

Jio itself. When a tower's signal drops, Jio's engineers have to decide: is it rain, or is something broken? If JioMausam can tell them before they send a repair team, Jio saves trips. Then JioMart can plan deliveries around rain, and the MyJio app can send rain alerts. If it helps inside Reliance, we sell the same alerts to other companies.

7 · What do other companies buy?

Not a pretty map. They buy a message like “Rain is starting in Zone A”. Then **they** decide what to do: move riders, change delivery times, or pay riders more to keep working.

8 · What about insurance (Mausam Kavach)?

Our first idea was insurance for workers: if heavy rain or extreme heat hits your area, you automatically get money, e.g. ₹300, without filling in forms. We still like it, but we found it is risky to start with: we don’t know who would pay, and beams alone are not accurate enough to pay people on. So insurance comes last, only if the system passes all accuracy checks, and a real insurance company (not Jio) carries the risk.

9 · How did we work out prices?

We counted, from 25 years of weather records, how many days a year each city gets extreme rain or heat. Fewer bad days means a cheaper price. Mumbai’s extreme-rain cover works out to about ₹45 a month, Nagpur’s heat cover ₹52, Ahmedabad’s ₹29. These are starting points; a real insurer would set the final price.

10 · What are we asking for?

Ninety days to answer three questions. Does Jio have enough beams? Does the team-up reach our accuracy target? Does it actually help a Reliance business? If all three say yes, grow it. If not, stop. That’s cheap, and nobody has to believe us; the test decides.

THE WHOLE THING IN ONE BREATH

Jio’s towers may already be able to sense rain. We checked the science, found its limit, fixed it by combining sources, and want to prove it inside Reliance before selling it. Insurance is a possible extra, not the main business.

Our two deliverables: the video and the presentation

Judges have seen our **video** (submitted earlier, 1 minute 45 seconds, it cannot be changed). In the round we give the **presentation** (10 slides, 3 minutes) and then answer questions for 4 minutes. The video was our first version; the presentation is the stress-tested version. Expect questions about the gap between them (Part 3).

The video, scene by scene

SCENE	ON SCREEN	SAID ALOUD
1 • Hook	"India's biggest weather network already exists." Lakhs of stations · every district · switched on 24×7 · "It's called Jio."	<i>"Nobody noticed, because we call it Jio."</i>
2 • Problem	Weather is hyperlocal, our data isn't · farmers, delivery riders, construction workers, street vendors · ~40 Doppler radars for 3.3 million km ² (IMD) · 34M full-time jobs' worth of work hours lost to heat by 2030 (ILO 2019) · months for a crop-insurance claim	<i>"...Heat alone could cost India thirty-four million jobs by 2030, and when insurance exists, claims take months."</i>
3 • Insight	Rain weakens the signal between towers (Messer et al., Science 2006) · phone batteries feel the city's heat, 41.8°C city average (Overeem et al., GRL 2013) · "Every tower is a rain gauge. Every phone is a thermometer."	<i>"We don't build sensors. We become the sensor."</i>
4 • Product	JioMausam: a live, street-level weather map of India built from network data, zero new hardware, illustrative 1 km grid · MyJio recharge: ₹299 + Mausam Kavach heat + rain cover ₹29/mo · "46.2°C near you, measured by the network in Nagpur. Kavach triggered. ₹300 credited to JioPay, paid 2 hours after the trigger."	<i>"When the network measures 45 degrees near you, the payout lands in your JioPay wallet within hours. No claim forms. No waiting."</i>
5 • Business	B2C Kavach ≈ ₹170 Cr/yr (1% of 500M users = 5M policies × ₹29/month), underwritten with JFS insurance partners · B2B cover + data: gig platforms, insurers, state disaster agencies, agri companies · customer zero: JioMart rainy-day demand, New Energy solar output, Jio rain-fade prediction	<i>"...about 170 crore rupees a year. Gig platforms and insurers buy the data. And Reliance is customer zero."</i>
6 • Moat	"Why only Reliance? Every layer. One roof." Sense (lakhs of sites) · Reach (500M+ subscribers, every recharge) · Pay (JioPay, JioFinance) · Insure (JFS partners) · Use (JioMart, Reliance New Energy)	<i>"It owns every layer... under one roof."</i>
7 • Roadmap and close	Days 0–90 pilot: rain in Vidarbha, heat in Ahmedabad, validated against IMD gauges · months 4–9 IRDAI sandbox with JFS partners · Year 2 national map, open API	<i>"India will adapt to climate change by listening to the infrastructure it already has. This is JioMausam. Different by design."</i>

The presentation, at a glance

#	SLIDE TITLE	WHY IT IS THERE
1	India has a weather-decision problem	Make the problem concrete in ten seconds: the data exists, but it is too coarse to act on zone by zone.
2	Jio's microwave links may carry a weather signal	The insight, and a precise definition of JioMausam.
3	Links alone are promising, but fall short of our bar	Credibility: show what is proven, what we checked ourselves, what is not proven, and the honest limit.

4	Let accuracy determine the mix	Answer the limit: combine sources, test every mix, keep the cheapest that clears the bar.
5	Reliance is the first proving ground	Remove the biggest commercial risk: prove value inside Reliance before selling outside.
6	Then sell the same intelligence to the market	External market, and what the product actually is.
7	The edge is integration, not the sensor	Why Jio, and how value is created without insurance risk.
8	Insurance is one application, not the anchor	Show the pivot from the video, and that insurance is gated.
9	If we offer Kavach, weather risk sets the starting price	Keep continuity with the video's pricing, and show prices come from data.
10	Don't believe us. Test it in 90 days	A concrete, low-risk ask with stop/scale points.

What we say in the presentation (about 3 minutes)

Slide 1. Imagine you're running deliveries in Mumbai. Your dashboard says: "Heavy rain in Mumbai." Do you stop the whole city? On 26 July 2005, Santacruz got 944 millimetres of rain. Colaba, 20 kilometres away, got 73. The problem isn't weather data. It's knowing what to do, zone by zone.

Slide 2. But Jio may already own infrastructure that can carry a signal about that rain. Rain weakens microwave signals between telecom sites. Measure the extra loss and you can estimate rain along each link. JioMausam combines that with radar and gauges into alerts businesses can act on.

Slide 3. Sounds great? We thought so too. The physics is well established, so we tested it ourselves: 500 real links from a public dataset in Germany, not Jio's. The links tracked radar closely, but alone they caught only 68% of rain events. Our bar is 80%. That gap told us how to build JioMausam.

Slide 4. So we don't bet on one sensor. Accuracy decides the mix: Jio's links, network-health data to filter out faults, radar, gauges, satellite. The pilot tests every combination against held-out gauges. The answer isn't "Jio only". It's the cheapest mix that works.

Slide 5. Before selling to Swiggy, where do we prove it? Inside Reliance, starting with Jio. A signal drops: is it rain, or is something broken? If JioMausam can answer that before a crew goes out, that's real value. Then JioMart tests commercial value; MyJio tests reach.

Slide 6. Once that works, the same intelligence goes outside. Rain hits Zone A. JioMausam flags it. The platform decides whether to reroute, delay or raise incentives. That's the product: not a prettier weather map, but a signal that changes a business decision.

Slide 7. So why Jio? Not because it invented the physics. The edge is integration: Jio's signal, JioMausam's intelligence, Reliance's proving ground and reach. We can monetise the intelligence outside and create savings inside Reliance, even if insurance never launches.

Slide 8. Kavach is where we started: protection for workers. Then we stress-tested it. Who pays? Is the trigger accurate? Can an insurer price it? The insurance isn't the business; the intelligence underneath it is. So Kavach becomes one application, only after the accuracy and actuarial gates pass.

Slide 9. If we offer it, Indian weather risk sets the starting point for the price. From 25 years of IMD records: about 45 rupees a month in Mumbai, 52 in Nagpur, 29 in Ahmedabad, and 208 for Mumbai Plus. All illustrative until actuarial validation.

Slide 10. We're not asking you to believe us. We're asking Jio to test it. Ninety days, three questions. Does Jio have the coverage? Does the combined system clear our bar? Does it improve a real Reliance operation? If yes, scale. If not, stop. First prove the signal. Then sell the intelligence. Protection comes last.

What we removed from the video, and what we refined

After the video we stress-tested three things: **accuracy** (our own check: links alone caught 68% of rain events), **who pays** (unproven) and **evidence** (several claims we couldn't back). We kept the core idea and changed what didn't survive.

THE LINE TO OWN THE CHANGE

"Our original idea was Kavach. After stress-testing the accuracy and who pays, we realised the stronger business is the weather intelligence underneath it. Kavach becomes one application of that intelligence."

THE VIDEO SAID

"India's biggest weather network already exists... lakhs of stations, every district."

THE PRESENTATION SAYS

"Jio may already own the infrastructure for a new weather-sensing layer."

Why: Only microwave links sense rain; fibre does not. We don't know how many of Jio's sites use usable links, so "lakhs" and "biggest" were claims we couldn't prove.

If asked: "The video was our first version. We learned only microwave links work, so how many Jio has is now Gate 1."

THE VIDEO SAID

"Every tower is a rain gauge." / "...tells you exactly how hard it's raining."

THE PRESENTATION SAYS

"Measure the excess loss and estimate path-averaged rainfall." Links alone caught 68% of rain events.

Why: A link gives an average over its whole path, not an exact point reading, and our own test showed links alone miss about a third of heavy-rain events.

If asked: "Our own check showed the limit, so we combine links with radar and gauges."

THE VIDEO SAID

"Phone batteries... track air temperature. Every phone is a thermometer." (41.8°C crowd-sourced)

THE PRESENTATION SAYS

Removed. Heat comes from IMD weather stations and satellite, through the same engine. Towers sense rain only.

Why: The research is real (Overeem et al., 2013) but it reads phone battery temperatures through an app, which needs users' consent under India's data-protection law; it gives city-wide daily averages, not street-level readings; and phone heat is distorted by charging, pockets and air-conditioning. Not good enough to pay money on.

If asked: "It's published science, but it needs personal phone data and only works as a city-wide daily average, so for heat we use IMD stations and satellite."

THE VIDEO SAID

"We don't build sensors. We become the sensor." / "Zero new hardware."

THE PRESENTATION SAYS

"The edge is integration, not the sensor." New gauges on Jio sites only where links are thin.

Why: If links are thin somewhere, a few cheap gauges on Jio sites are the backup. Promising zero hardware would remove our safety net.

If asked: "Mostly existing infrastructure; a few gauges only where links are thin."

THE VIDEO SAID

"A live, street-level weather map of India... illustrative 1 km grid."

THE PRESENTATION SAYS

"Decision infrastructure, not a weather map"; "spatial weather grid"; near-real-time.

Why: Resolution depends on how dense the links are, and refresh speed depends on Jio's data systems; both are pilot questions. Businesses pay for decisions, not maps.

If asked: "We don't promise 1 km everywhere; the pilot finds the real resolution."

THE VIDEO SAID

"When the network measures 45 degrees near you..." / "46.2°C near you, measured by the network in Nagpur."

THE PRESENTATION SAYS

Heat trigger 47°C, from IMD stations and satellite.

Why: The network doesn't measure temperature. And 45°C happens about 10 days a year in Nagpur, which would price cover at about ₹168 a month; 47°C (about 2 days a year) gives ₹52.

If asked: "The phone mock-up was illustrative. Real heat data comes from IMD; 47°C keeps the price affordable."

THE VIDEO SAID

"Mausam Kavach... insurance on your recharge for ₹29 a month" (heat + rain, everywhere).

THE PRESENTATION SAYS

City-specific prices: Mumbai rain ₹45, Nagpur heat ₹52, Ahmedabad heat ₹29 (minimum), Mumbai Plus ₹208. Recharge is a Year-2 option.

Why: 25 years of IMD records show risk differs a lot by city. A flat ₹29 would lose money in Mumbai and Nagpur. ₹29 survives as our minimum price.

If asked: "When we priced it with real IMD data, one flat price didn't fit; ₹29 became the minimum."

THE VIDEO SAID

"The payout lands in your JioPay wallet within hours" / "₹300 credited, paid 2 hours after the trigger."

THE PRESENTATION SAYS

"A licensed insurer would underwrite and pay." ₹300 is a prototype assumption.

Why: The insurer pays through its own rail (UPI or bank); nothing depends on JioPay. Speed is measured in shadow mode, not promised.

If asked: "The insurer pays; we'd measure speed in shadow mode first."

THE VIDEO SAID

"Underwritten with Jio Financial Services' insurance partners." / "Months 4-9: IRDAI sandbox, Kavach launched with JFS partners."

THE PRESENTATION SAYS

"A licensed insurer would underwrite." Sandbox in months 4-12, only after the gates pass.

Why: We have no insurer partner, and JFS is a separately listed group company. Insurance launches only after accuracy and actuarial checks.

If asked: "JFS is a natural partner, but any deal would be at arm's length, and only after the gates."

THE VIDEO SAID

"One percent of Jio's users is five million policies, about ₹170 crore a year."

THE PRESENTATION SAYS

Removed from the deck. Q&A only, as an illustrative premium pool.

Why: Premiums belong to the insurer, not Jio; 1% uptake is a guess; and big untested numbers invite attack. (Maths: 5 million × ₹29 × 12 = ₹174 crore, rounded to ₹170.)

If asked: "That was the insurer's premium pool on a guessed uptake, not Jio revenue. We removed it."

THE VIDEO SAID

Insurance-first business ("three layers of revenue").

THE PRESENTATION SAYS

Intelligence first. Value = intelligence fees (external) + savings inside Reliance (internal) + distribution fees (optional).

Why: Stress-testing showed who pays for insurance is unproven, and the weather intelligence underneath is the stronger business.

If asked: "Our original idea was Kavach. Stress-testing showed the stronger business is the intelligence underneath it."

THE VIDEO SAID

"Reliance is customer zero: JioMart rainy-day demand, New Energy solar output, Jio rain-fade prediction."

THE PRESENTATION SAYS

"Reliance is the first proving ground": Jio network ops (hero), JioMart, MyJio.

Why: "Customer" implies Reliance has agreed to buy. Solar output needs sunshine and cloud data, which rain links don't measure. "Prediction" became detection: telling rain from faults is testable.

If asked: "We kept the uses our data can actually serve and turned them into tests."

THE VIDEO SAID

"Why only Reliance? It owns every layer... under one roof." (Sense, Reach, Pay, Insure, Use)

THE PRESENTATION SAYS

"The edge is integration, not the sensor": Sense → Understand → Reach → Protect.

Why: "Only" is easy to attack (Airtel has towers; weather firms have radar). Integration is defensible. "Pay" was dropped because the insurer pays.

If asked: "Others could do parts of this; Reliance can connect all the layers."

THE VIDEO SAID

"Weather is hyperlocal, our data isn't" + ~40 radars for 3.3 million km².

THE PRESENTATION SAYS

"India has a weather-decision problem" + the 944 vs 73 mm Mumbai example. Radar is an input we combine.

Why: Criticising radar invites "then just use radar". A concrete Mumbai example proves the problem better than a national statistic.

If asked: "Radar is part of our mix; the problem is turning observations into zone decisions."

THE VIDEO SAID

"Heat alone could cost India 34 million jobs by 2030."

THE PRESENTATION SAYS

Removed.

Why: The ILO figure is the *equivalent* of 34 million full-time jobs in lost working hours, not 34 million jobs lost; the spoken line overstated it. And heat is no longer our lead.

If asked: "The ILO estimated heat could cost India working hours equal to about 34 million full-time jobs by 2030."

THE VIDEO SAID

Farmers, construction workers, street vendors, "months for a crop-insurance claim".

THE PRESENTATION SAYS

Businesses first (delivery, then logistics and cities, then agriculture and others); workers through optional Kavach.

Why: A focused first market beats helping everyone at once. Agriculture stays in tier 3.

If asked: "We start where decisions happen many times a day: delivery."

THE VIDEO SAID

"Days 0–90: rain in Vidarbha, heat in Ahmedabad, validated against IMD gauges."

THE PRESENTATION SAYS

Three gates (coverage, accuracy, value). The coverage audit picks the city; back-test on archived Mumbai monsoon data; heat in months 4–12.

Why: Timing: a pilot starting now would miss Vidarbha's monsoon (June–September) and Ahmedabad's heat season (April–June). And we can't pick a city before knowing its coverage.

If asked: "A 90-day window now has neither monsoon in Vidarbha nor heat in Ahmedabad, so we back-test archived data and let the audit pick the live city."

THE VIDEO SAID

"Year 2: national map, open API."

THE PRESENTATION SAYS

Year 2: cities + enterprises; Year 3: national weather API. Only after all three gates pass.

Why: Scaling depends on coverage and calibration city by city.

If asked: "City by city, only where coverage supports it."

KEPT FROM THE VIDEO

- Jio's network as a weather signal, backed by Messer et al., Science 2006
- The names JioMausam and Mausam Kavach, and covering both rain and heat
- ₹300 fixed benefit as the prototype payout, "no claim forms, no surveyor"
- Selling data and alerts to businesses (platforms, insurers, cities, agriculture)
- Using Reliance's own businesses first
- A 90-day pilot, an IRDAI sandbox stage, and a national API later
- The closing idea: listen to infrastructure India already has · "Different by design"

NEW IN THE PRESENTATION

- The Mumbai 944 vs 73 mm example
- Our own test on 500 real links, with the honest 68% result
- Combining sources, with accuracy gates and held-out gauges
- Network-health data to tell rain from faults
- Network operations as the hero internal use
- The "customer decides" chain: Jio sells alerts, businesses decide
- City-specific prices from 25 years of IMD records
- Explicit stop-or-scale gates

IF THEY ASK “WHY IS YOUR PRESENTATION SO DIFFERENT FROM YOUR VIDEO?”

“The video was our first version. Afterwards we stress-tested accuracy, who pays and evidence. Our own test showed links alone catch 68% of rain events, not enough to pay claims on, and we couldn’t prove who would pay for insurance. So we kept the core, Jio’s network as a weather signal, and moved insurance from the anchor to an option. We’d rather show you what survived testing than defend what didn’t.”

The presentation, slide by slide

For each slide: why it exists, what is on screen, what you say, where every number comes from, and the trap to avoid. Labels: **FACT** **OUR CALCULATION** **OUR TARGET** **OUR ASSUMPTION** **PLAN**.

SLIDE 1 India has a weather-decision problem

Why it's there: Make the problem concrete in ten seconds: the data exists, but it is too coarse to act on zone by zone.

ON SCREEN

- Bar chart: Santacruz 944 mm vs Colaba 73 mm, 26 July 2005, stations ~20 km apart (IMD)
- "What a city-level alert says: Heavy rain in Mumbai" → "What it needs to decide: Which zones do I act on, right now?"
- Businesses decide blind · Workers carry the loss

WHAT YOU SAY

Imagine you're running deliveries in Mumbai. Your dashboard says: "Heavy rain in Mumbai." Do you stop the whole city? On 26 July 2005, Santacruz got 944 millimetres of rain. Colaba, 20 kilometres away, got 73. The problem isn't weather data. It's knowing what to do, zone by zone.

NUMBERS AND WHERE THEY COME FROM

944 vs 73 mm

FACT

IMD rainfall records

Watch out: Present the alert as imagined ("Imagine you're running deliveries..."), not as something a named business receives. If they say "one day proves nothing", use the 2020–24 station comparison (Part C, "The problem and the idea").

SLIDE 2 Jio's microwave links may carry a weather signal

Why it's there: The insight, and a precise definition of JioMausam.

ON SCREEN

- Two towers and a microwave link; rain weakens the signal; measure the excess loss → path-averaged rainfall, near-real-time
- "This technique uses microwave links, not fibre."
- JioMausam is decision infrastructure, not a weather map: link signals + IMD radar + gauges → spatial weather grid → decision-ready alerts and triggers
- It is / It isn't (not a new sensor network, not a consumer weather app, not an insurer, no heat sensing from towers)
- Takeaway: "Jio may already own the infrastructure for a new weather-sensing layer."

WHAT YOU SAY

But Jio may already own infrastructure that can carry a signal about that rain. Rain weakens microwave signals between telecom sites. Measure the extra loss and you can estimate rain along each link. JioMausam combines that with radar and gauges into alerts businesses can act on.

Watch out: Keep "may". Never say "every tower is a rain gauge" or that Jio can already sense the weather.

SLIDE 3 Links alone are promising, but fall short of our bar

Why it's there: Credibility: show what is proven, what we checked ourselves, what is not proven, and the honest limit.

ON SCREEN

- Proven · published: the physics; Science 2006; PNAS 2013 (rain mapped across a whole country)
- Our check · public data: we ran the open-source method on 500 real links with radar as reference
- Not yet proven · the pilot: Jio's coverage, Indian accuracy, business value, who pays, insurance-grade triggers
- Banner: OUR CHECK · 500 REAL LINKS · PUBLIC RESEARCH DATA, GERMANY · NOT JIO DATA
- 68% of rain events caught by links alone, against our 80% bar; event = 10 mm+ in 3 hours in a ~4 km zone; 8 unseen test days
- Chart: network average, hour by hour, links vs radar

WHAT YOU SAY

Sounds great? We thought so too. The physics is well established, so we tested it ourselves: 500 real links from a public dataset in Germany, not Jio's. The links tracked radar closely, but alone they caught only 68% of rain events. Our bar is 80%. That gap told us how to build JioMausam.

NUMBERS AND WHERE THEY COME FROM

68%	OUR CALCULATION	pycomlink public dataset (KIT), German radar
80%	OUR TARGET	Pilot go/no-go bar
10 mm in 3 h	OUR ASSUMPTION	Event rule; 5 and 15 mm also tested
Science 2006, PNAS 2013	FACT	Published papers

Watch out: Never say "studies show 68%" (it is ours) or "we tested Jio's network". Mention 0.95 only if asked.

SLIDE 4 Let accuracy determine the mix

Why it's there: Answer the limit: combine sources, test every mix, keep the cheapest that clears the bar.

ON SCREEN

- Sources and providers: Jio microwave links (Jio-owned) · network health: faults, outages, maintenance (Jio-owned) · Doppler radar (IMD, govt) · rain gauges (IMD · BMC) · satellite (External) · new gauges on Jio sites (New install) · commercial data (Commercial)
- Mix bars scored against held-out rain gauges: Links only = our check 68%; other mixes “measured in the pilot”; “Radar + gauges, no links” shows what Jio adds
- Rule: the lowest-cost mix that clears every accuracy gate ($\geq 80\%$ of predefined trigger events, totals within $\pm 10\%$, under 15% unconfirmed)

WHAT YOU SAY

So we don't bet on one sensor. Accuracy decides the mix: Jio's links, network-health data to filter out faults, radar, gauges, satellite. The pilot tests every combination against held-out gauges. The answer isn't “Jio only”. It's the cheapest mix that works.

NUMBERS AND WHERE THEY COME FROM

68%	OUR CALCULATION	Same as slide 3
80%, $\pm 10\%$, <15%	OUR TARGETS	Pilot gates

Watch out: The pilot does the scoring; we have not done it yet. No prices are claimed; labels show who provides each source.

SLIDE 5 Reliance is the first proving ground

Why it's there: Remove the biggest commercial risk: prove value inside Reliance before selling outside.

ON SCREEN

- Hero: Jio network operations. Can we tell rain-related signal loss from equipment faults before sending a crew? Potentially fewer wasted visits and earlier flood alerts. Jio owns the network data and controls the operational decision.
- “Before Jio sells weather intelligence to anyone, it uses it to understand its own network.”
- JioMart and Reliance Retail: rain-aware dispatch, delivery promises, zone pauses (tests commercial value)
- MyJio: street-level rain alerts (tests reach at scale)

WHAT YOU SAY

Before selling to Swiggy, where do we prove it? Inside Reliance, starting with Jio. A signal drops: is it rain, or is something broken? If JioMausam can answer that before a crew goes out, that's real value. Then JioMart tests commercial value; MyJio tests reach.

Watch out: Everything here is a hypothesis to test, not a result. Say “proving ground”, never “first customer”.

SLIDE 6 Then sell the same intelligence to the market

Why it's there: External market, and what the product actually is.

ON SCREEN

- Tiers: 1 delivery, quick commerce, ride-hailing (Alerts) · 2 logistics, cities and transport, insurers (Alerts + data) · 3 agriculture, power, construction, retail (Data + alerts)
- Example chain: Rain hits Zone A → JioMausam sends zone alert + rain estimate → the customer decides (dispatch, ETAs, incentives) → lower disruption
- Why buy from us: dense sensing wherever usable links exist · potentially denser than gauges · near-real-time monitoring · potentially valuable wherever microwave backhaul is present · auditable trigger from multiple sources

WHAT YOU SAY

Once that works, the same intelligence goes outside. Rain hits Zone A. JioMausam flags it. The platform decides whether to reroute, delay or raise incentives. That's the product: not a prettier weather map, but a signal that changes a business decision.

Watch out: Company names are examples of buyers, not customers we have spoken to. Jio sells intelligence; the customer owns the decision.

SLIDE 7 The edge is integration, not the sensor

Why it's there: Why Jio, and how value is created without insurance risk.

ON SCREEN

- Sense (Jio network) → Understand (JioMausam) → Reach (Reliance ecosystem) → Protect (JFS + licensed insurers, optional)
- "Reliance can connect all four layers, so Jio earns from the intelligence without carrying insurance risk."
- How Jio creates value: O1 external revenue, intelligence fees · O2 external, optional, distribution fees · O3 internal value, savings inside Reliance

WHAT YOU SAY

So why Jio? Not because it invented the physics. The edge is integration: Jio's signal, JioMausam's intelligence, Reliance's proving ground and reach. We can monetise the intelligence outside and create savings inside Reliance, even if insurance never launches.

Watch out: No claims about competitors. JFS is a separately listed group company; any deal would be at arm's length.

SLIDE 8 Insurance is one application, not the anchor

Why it's there: Show the pivot from the video, and that insurance is gated.

ON SCREEN

- Flow: qualifying weather event in a zone → JioMausam supplies the trigger data → a licensed insurer would underwrite and pay → fixed benefit to the worker, e.g. ₹300
- Eligible = already active in the zone before the event began (predefined rules); ₹300 is a prototype assumption
- Why optional: only after accuracy and actuarial gates (shadow payouts until then) · who pays is unproven · the insurer carries the risk, never Jio · if it fails, nothing breaks

WHAT YOU SAY

Kavach is where we started: protection for workers. Then we stress-tested it. Who pays? Is the trigger accurate? Can an insurer price it? The insurance isn't the business; the intelligence underneath it is. So Kavach becomes one application, only after the accuracy and actuarial gates pass.

NUMBERS AND WHERE THEY COME FROM

₹300

OUR ASSUMPTION

Prototype benefit

Watch out: No insurer partner yet: always "would". Kicker reads "Where we started": own the pivot.

SLIDE 9 If we offer Kavach, weather risk sets the starting price

Why it's there: Keep continuity with the video's pricing, and show prices come from data.

ON SCREEN

- Table: Mumbai Basic ≥ 204.5 mm/day (1.1 days → ₹45) · Nagpur $\geq 47^\circ\text{C}$ (2.2 raw, ~1.2 paid → ₹52) · Ahmedabad $\geq 47^\circ\text{C}$ (0.7, ~0.6 → ₹29 floor) · Mumbai Plus ≥ 64.5 mm/day (11.9 raw, 5 paid → ₹208)
- Price = expected paid days $\times$ ₹300 $\div$ 60% $\div$ 12
- City cards: Mumbai 11.9 heavy-rain days a year (Phase 1) · Nagpur 2.2 and Ahmedabad 0.7 days at 47°C + (Phase 2)
- Footer: ₹300 and 60% are prototype assumptions; 5-day annual cap; final pricing subject to actuarial validation

WHAT YOU SAY

If we offer it, Indian weather risk sets the starting point for the price. From 25 years of IMD records: about 45 rupees a month in Mumbai, 52 in Nagpur, 29 in Ahmedabad, and 208 for Mumbai Plus. All illustrative until actuarial validation.

NUMBERS AND WHERE THEY COME FROM

Trigger levels 204.5 / 64.5 mm

FACT

IMD rain categories

47°C

OUR ASSUMPTION

Rare heat level

Days a year, paid days

OUR CALCULATION

Counted from 25 years of IMD station reports

Prices

OUR CALCULATION

Formula above

Watch out: ₹208 is not priced from the 10 mm-in-3-hours rule. All prices are illustrative.

SLIDE 10 Don't believe us. Test it in 90 days

Why it's there: A concrete, low-risk ask with stop/scale points.

ON SCREEN

- Days 1–30 Coverage (Gate 1: enough usable coverage?) · Days 15–60 Accuracy (Gate 2: fused system clears every gate?) · Days 61–90 Value (Gate 3: improves a real Reliance operation? matched zones with and without JioMausam)
- Roadmap: 0–3 m three gates / shadow payouts · 4–12 m Reliance live + first clients / sandbox + heat · Year 2 cities + enterprises / Basic at recharge · Year 3 national weather API / national scale
- “Only after all three gates pass do we scale externally.”
- Close: “We aren’t asking Jio to build another weather network...”; “First prove the signal. Then sell the intelligence. Protection comes last.”

WHAT YOU SAY

We're not asking you to believe us. We're asking Jio to test it. Ninety days, three questions. Does Jio have the coverage? Does the combined system clear our bar? Does it improve a real Reliance operation? If yes, scale. If not, stop. First prove the signal. Then sell the intelligence. Protection comes last.

NUMBERS AND WHERE THEY COME FROM

90 days, gates

PLAN / TARGETS

Our pilot design

Watch out: “We don’t pre-select the city; the coverage audit does.”

How we tested the microwave links: every detail

The full chain behind “links alone caught 68%”: the data, the tools, every step with a real link, the scores, the map and the decision test. Every number here can be rechecked from public data.

5.1 The question we asked

Can the signal between two phone towers tell us how much it is raining, and is it good enough to make a yes/no decision (“pay” or “alert”) for a neighbourhood? We did **not** use Jio data (it isn’t public). We used the standard public research dataset that has both real tower links and an answer key.

5.2 The data: what, where from, how big

FILE	WHAT IS INSIDE	SIZE
<code>example_cml_data.nc</code>	500 real microwave links from a German mobile network. Each link has 2 channels (one per direction). For every minute from 10 to 20 May 2018: tsl (power sent, dBm) and rsl (power received, dBm). Per link: frequency, polarization, length, tower coordinates.	500 links × 2 channels × 15,840 minutes
<code>example_path_averaged_reference_data.nc</code>	German Weather Service radar rain averaged along each link’s path, every 5 minutes. The answer key for each link.	500 links × 3,168 five-minute steps
<code>example_areal_reference_data.nc</code>	The radar rain map itself (product “RADOLAN YW”, Deutscher Wetterdienst), every 5 minutes. The answer key for the map and the decision test.	190 × 228 squares of ~1 km

- **Where from:** the example data shipped with **pycomlink**, the open-source toolbox from Karlsruhe Institute of Technology (KIT), Germany, BSD licence: github.com/pycomlink/pycomlink, folder `pycomlink/io/example_data` (also installs with `pip install pycomlink`). Our copies are byte-for-byte identical.
- **The links:** length 0.5–28.6 km (typical 5.8 km); frequency 6.5–38.9 GHz (typical 24.9 GHz); 450 vertical and 50 horizontal polarization.
- **Location:** real network, but coordinates were shifted for anonymity (they appear in the North Sea). The pattern is real; the place is not.

5.3 The tools

No AI and no machine learning: physics formulas plus simple statistics, in Python (xarray, numpy, scipy). From **pycomlink 0.6.0** we used three standard functions: the dry baseline, the wet-antenna correction (Schleiss et al., 2013) and the rain formula (ITU-R P.838, 2005 coefficients). Our own code does everything else: cleaning, wet/dry detection, calibration, the map and the decision test. File: `prototype/pipeline/rain_pipeline.py`. The Test Lab (a web page) repeats every step in your browser, and the Excel workbook recomputes the key numbers with live formulas.

5.4 The physics in one page

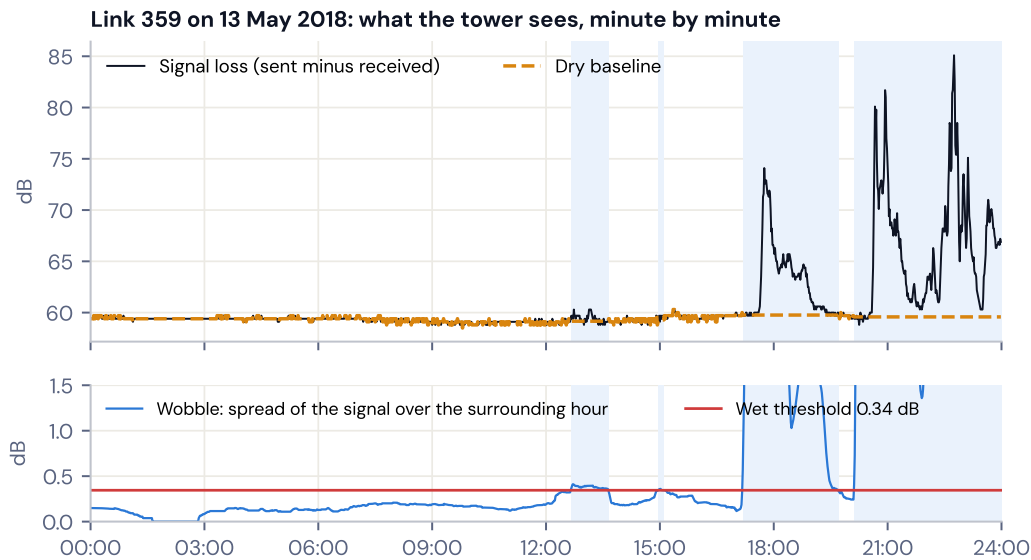
- **dB and dBm.** Radio power is measured on a log scale. dBm is power compared with one milliwatt. A difference of 3 dB means half the power; 10 dB means one-tenth.
- **Signal loss** = power sent – power received. On a dry day it is steady (for link 359 about 59 dB). Rain adds extra loss.

- **The formula** (ITU-R P.838): $k = a \times R^b$. k = extra loss per kilometre (dB/km), R = rain rate (mm per hour), a and b = constants that depend on frequency and polarization. Turn it round: $R = (k \div a)^{1/b}$.
- **Frequency matters.** Higher frequencies react more to rain (a is bigger). Very low-frequency, very long links barely react, which is why they score worse.
- **Path-averaged.** The loss happens along the whole beam, so a link tells you the average rain along its path, not the rain at one point.

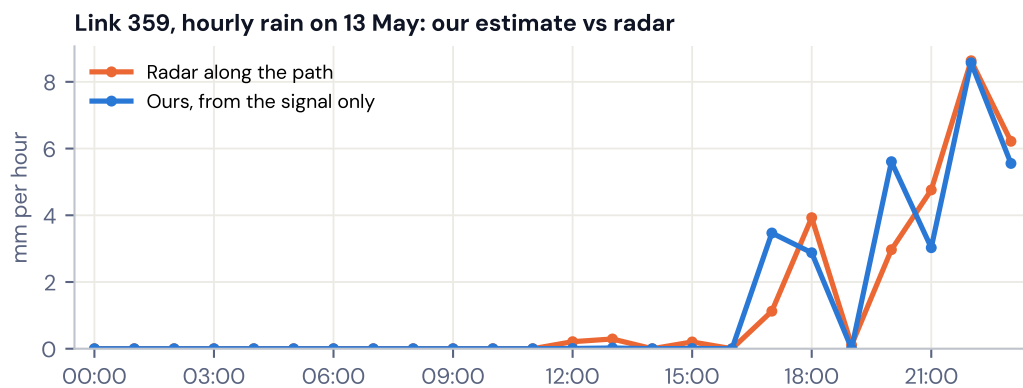
5.5 Step by step, with a real link

Link 359: 8.47 km long, 23.1 GHz, vertical polarization ($a = 0.129$, $b = 0.962$). We compare a dry minute with a storm minute on 13 May 2018, 22:45.

#	STEP	DRY MINUTE	STORM MINUTE	NOTES
1	Clean. Remove error codes (tsl 255, rsl -99.9); fill gaps of up to 5 minutes with a straight line			Gaps longer than 5 min stay empty
2	Signal loss = sent – received	$19.0 - (-40.4) = 59.4 \text{ dB}$	$19.0 - (-66.1) = 85.1 \text{ dB}$	
3	Wet or dry? Standard deviation (“wobble”) of the loss over the surrounding 60 minutes. Wet if above $1.12 \times$ that channel’s 80th-percentile wobble	0.11 dB → dry	5.91 dB → wet	Link 359’s threshold: 0.345 dB. 453 of 1,440 minutes that day were wet
4	Dry baseline. While dry, baseline = the loss itself; when wet, frozen at the average of the last 5 dry minutes	59.4	59.58 dB	The “no-rain” level
5	Wet antenna. Water on the antennas adds loss that isn’t rain in the path; allowed up to 1.5 dB, building up over ~15 minutes	0	1.50 dB	Schleiss et al. 2013
6	Rain loss $A = \text{loss} - \text{baseline} - \text{wet antenna}$ (negative → 0)	0	$85.1 - 59.58 - 1.50 = 24.02 \text{ dB}$	
7	Loss per km $k = A \div \text{length}$	0	$24.02 \div 8.47 = 2.84 \text{ dB/km}$	
8	Rain rate $R = (k \div a)^{1/b}$; below 0.1 mm/h → 0	0	$(2.84 \div 0.129)^{1/0.962} = 24.7 \text{ mm/h}$	ITU-R P.838
9	Both channels averaged	0	24.1 mm/h	One number per link per minute
10	Hourly: average each hour’s 60 minutes (= mm fallen in that hour)			Radar: 5-min amounts $\times 12 \rightarrow \text{mm/h}$, then hourly



Link 359 on 13 May 2018, minute by minute. Top: signal loss (black) and the dry baseline held during rain (dashed). Bottom: the wobble and the wet threshold (0.34 dB). Blue bands: minutes marked wet (453 of 1,440).



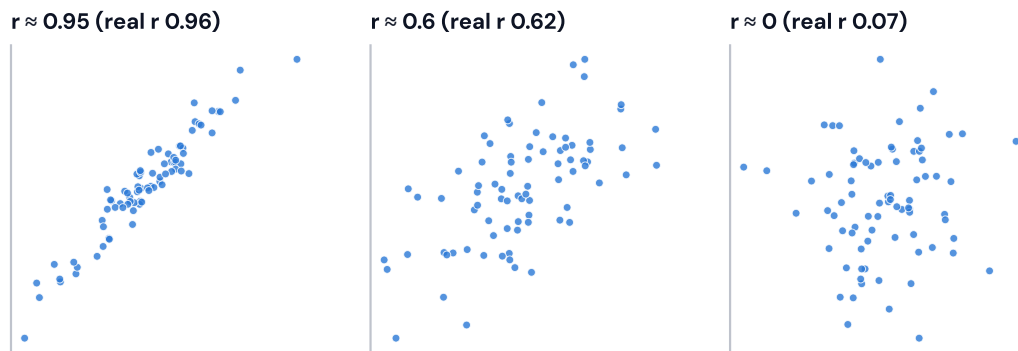
Link 359's hourly rain on 13 May: our estimate from the signal alone vs radar along the same path. Day total: ours 29.1 mm, radar 28.5 mm.

5.6 Calibration: one number, fitted fairly

Raw link estimates come out a bit low overall, so we fit **one scale factor K for the whole network**, using only the first three days (10–12 May, the “tuning days”): $K = \text{radar's total} \div \text{our total} = 2,758 \text{ mm} \div 2,352 \text{ mm} = 1.17$. Every estimate is multiplied by 1.17. The last eight days (13–20 May, 192 hours) were **never** used for tuning, so every score below is on unseen data. It is like studying from last year's paper and then sitting a new exam.

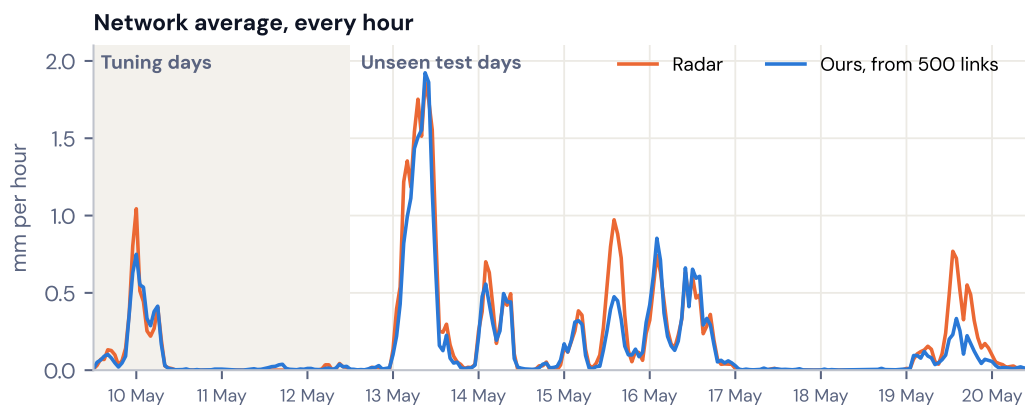
5.7 Scoring the rain amounts (unseen days only)

Correlation (r) measures whether two series rise and fall together: 1 = perfectly together, 0 = no relation. It does **not** check amounts, so we also report **bias** (our total $\div$ radar's total).

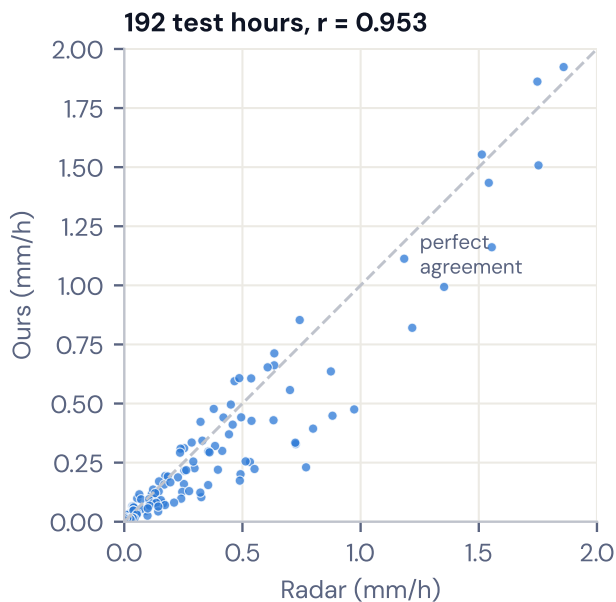


Made-up example dots showing what different correlations look like.

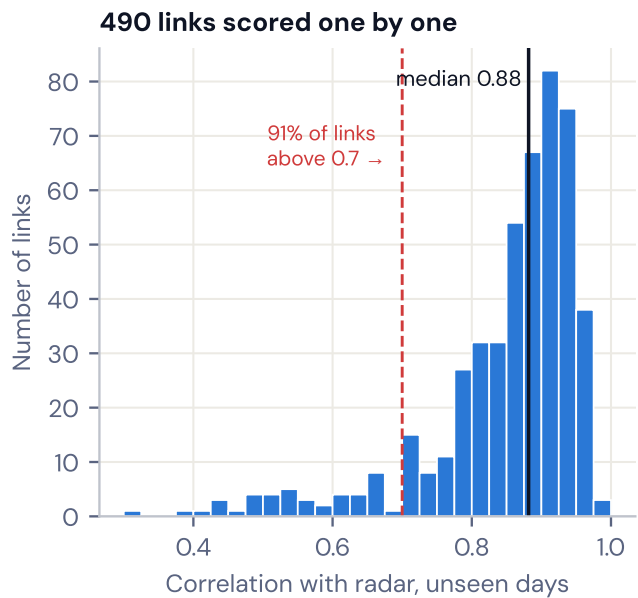
SCORE	HOW IT'S CALCULATED	WHAT IT MEANS
0.953 network	Average all links each hour; correlate with radar's average over 192 hours	The network as a whole tracks rain. Most forgiving: local errors cancel
0.882 typical link	Each link vs radar along its own path; middle value of 490 links (those with ≥ 100 valid hours and ≥ 5 mm of radar rain)	91% of links score above 0.7. Range 0.16–0.99
0.801 pooled	All link-hours thrown together	Mixes good and weak links and quiet hours
0.779 map	Our map vs radar's map, every square, every test hour	Adds map-making error; lowest
0.82 bias	Our total ÷ radar's total on test days (17,438 ÷ 21,312 mm)	We read 18% low; likely because the tuning days had little rain
91% / 5%	Per link per day: rain or no rain (0.1 mm+)	Rain days spotted 91% of the time, 5% false calls



Average rain over all 500 links, every hour for 11 days. Shaded: tuning days. Two completely different instruments: tower signals and weather radar.



Each dot is one test hour. Dots on the dashed line would be perfect agreement; they sit tightly along it, a little below (the 18% under-reading).



Each link scored on its own. Most sit between 0.8 and 0.95.

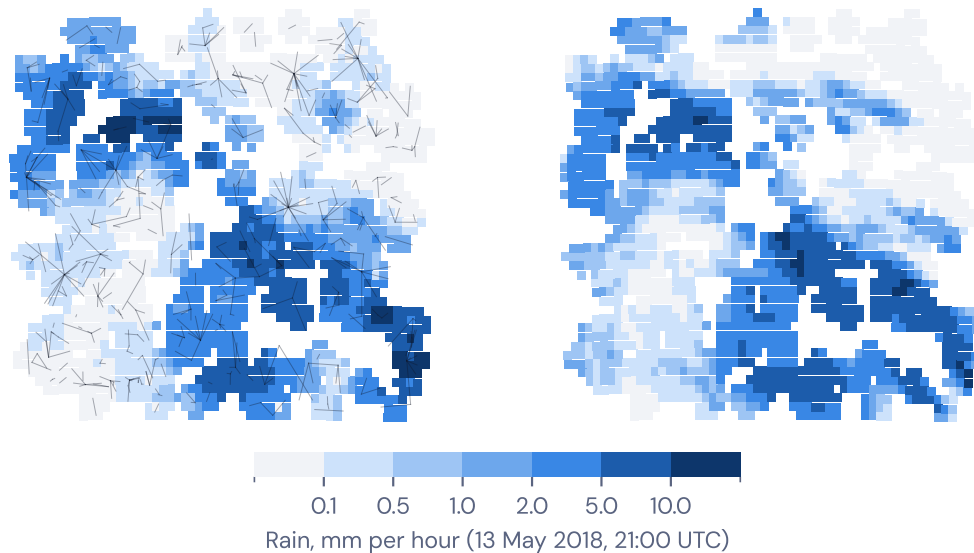
EXAMPLE LINK	LENGTH	FREQUENCY	R (UNSEEN)	WHY
Best · 118	2.6 km	37.4 GHz	0.99	Short path at a high frequency that reacts strongly to rain
Typical · 359	8.5 km	23.1 GHz	0.88	The example above
Weak · 33	28.6 km	6.5 GHz	0.70	Very long, low-frequency link; reacts weakly, so small errors grow (it read 165 mm vs radar's 78)

5.8 Making a map

1. Radar's ~1 km map is merged 4 × 4 into squares of about 3 × 4 km ("~4 km zones").
2. Each link's hourly rain is placed at the midpoint of its path.
3. Each square takes the **8 nearest midpoints within 20 km** and averages them, nearer links counting more: weight = $1 \div \text{distance}^2$ (a link 2 km away counts four times as much as one 4 km away). This is inverse distance weighting (IDW).
4. Only squares with a link midpoint within 6 km are kept: **1,782 squares**.

Our map, from 500 links only

Radar (the answer key)



The heaviest test hour (13 May, 21:00 UTC). Left: built only from the 500 links (thin lines). Right: radar. Squares are about 3×4 km.

5.9 The decision test: where 68% comes from

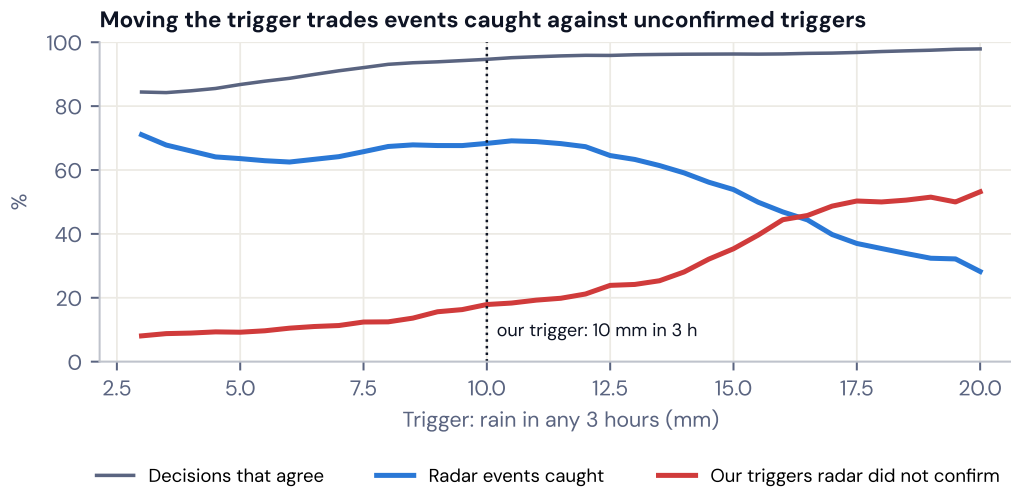
Rule: a square “triggers” on a day if any 3 consecutive hours that day add up to **10 mm or more**. We ask our map and radar’s map the same question for every square on every test day: 1,782 squares $\times$ 8 days = **14,256 yes/no decisions**.

	Radar: yes	Radar: no	Total
Links: yes	1,120 both yes	244 only us (unconfirmed)	1,364
Links: no	519 we missed	12,373 both no	12,892
Total	1,639	12,617	14,256

- **Events caught** = $1,120 \div 1,639 = 68.3\%$ (the number on slide 3).
- **Our triggers not confirmed by radar** = $244 \div 1,364 = 17.9\%$.
- **Decisions that agree** = $(1,120 + 12,373) \div 14,256 = 94.6\%$: looks great only because most square-days are dry; never headline it.

TRIGGER LEVEL	RADAR EVENTS	CAUGHT	UNCONFIRMED	AGREE
5 mm in 3 h	4,400	63.6%	9.2%	86.8%
10 mm in 3 h (ours)	1,639	68.3%	17.9%	94.6%
15 mm in 3 h	696	53.9%	35.3%	96.3%

No trigger level gets near 80%, so the conclusion doesn’t depend on the rule we picked.



Every trigger level from 3 to 20 mm re-scored. Lower triggers catch more events but agreement falls; higher triggers catch fewer and raise unconfirmed triggers.

5.10 Why links alone miss events

- They read 18% low overall, so events just over 10 mm fall just under.
- A link averages rain over its path, which smooths out sharp, local peaks.
- Squares between links are estimated from neighbours, not measured.
- One scale factor for 500 different links; production systems calibrate each area.
- A threshold test is harsh: 9 mm against an 11 mm event counts as a full miss.

THE 25-SECOND ANSWER

“We used a public research dataset from KIT in Germany: 500 real phone-network links, signal every minute for 11 days, with the German weather service’s radar as the answer key. Rain weakens the signal; a standard formula turns that loss into rain. For every ~4 km zone each test day we asked: 10 mm in 3 hours? Radar said yes 1,639 times; links also said yes on 1,120. That’s 68%, below our 80% bar, which is why we never use links alone.”

5.11 What this test proves, and what it doesn’t

IT SHOWS

Real phone-network links carry a strong rain signal (0.95 network, 0.88 typical link), with an open, published method that we ran ourselves.

IT DOESN’T SHOW

Anything about Jio’s network, Indian monsoon rain, rain gauges as truth (radar was the reference), or business value. Eleven days in May in Germany is a check of the method, not a validation of the product.

5.12 Check it yourself

- **Excel:** *data/JioMausam_Numbers_Workbook.xlsx* recomputes 1.17, 0.95, 0.88, 94.6% / 68% / 18% with live formulas from the CSVs.
- **CSV files** in *data/germany/csv/*: *links.csv*, *three_links_minute_by_minute.csv* (every step for links 118, 359, 33), *hourly_rain_for_all_links.csv*, *network_average_hourly.csv* (CORREL of the test rows = 0.953), *payout_decisions_test_days.csv* (the 14,256 decisions).
- **Test Lab and code:** *prototype/JioMausam_Test_Lab_standalone.html* reruns everything offline in a browser; the Python is *prototype/pipeline/rain_pipeline.py*.

The India data and pricing: every detail

Where the prices on slide 9 come from: real IMD records, simple counting, one formula, and a stress test of every year since 2000.

6.1 The data

- **Source:** NOAA's Global Summary of the Day, the public archive of daily reports from IMD's weather (synoptic) stations: noaa-gsod-pds.s3.amazonaws.com/{year}/{station}.csv. Public domain.
- **Stations:** Mumbai Santacruz 43003099999 · Mumbai Colaba 43057099999 · Nagpur Sonegaon 42867099999 · Ahmedabad airport 42647099999. Years 2000–2024.
- **Fields:** daily rainfall (inches × 25.4 = mm) and daily maximum temperature ((°F – 32) × 5/9 = °C). Codes 99.99 and 9999.9 mean missing.
- **Quality rule:** keep a year only if it has ≥330 days of reports and ≥110 in June–September. All 25 years passed for every city: 27,103 days in total.
- **Caveat:** a missing day counts as “no trigger”, so if anything we slightly undercount events.

6.2 The triggers

TRIGGER	WHERE IT COMES FROM
≥64.5 mm in a day	IMD's "heavy rain" category
≥115.6 mm in a day	IMD's "very heavy rain" category
≥204.5 mm in a day	IMD's "extremely heavy rain" category
≥47°C maximum	Our choice of a rare-heat level (45°C, close to IMD's heatwave level, happens too often to insure cheaply)

Median yearly rainfall for context: Mumbai 2,666 mm · Nagpur 1,047 mm · Ahmedabad 859 mm.

6.3 Counting days and the price formula

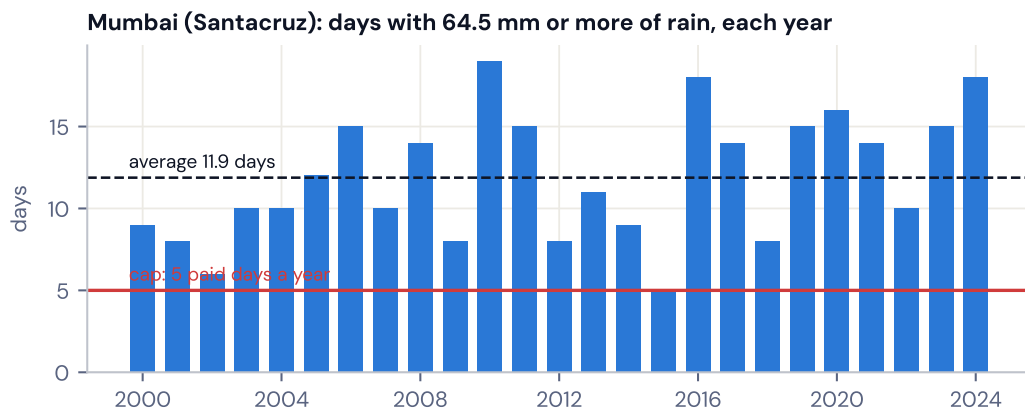
For each year, count the days at or above the trigger. Paid days are capped at 5 a year. Average over 25 years = expected paid days. Then:

$$\text{Monthly price} = \text{expected paid days} \times ₹300 \div 0.60 \div 12$$

- **₹300** = the fixed payout per trigger day (prototype assumption, about part of a day's earnings).
- **0.60** = target loss ratio: of every ₹100 of premium, ₹60 goes back as payouts, ₹40 covers the insurer's costs and margin (assumption).
- **÷ 12** turns a yearly premium into a monthly one.

PRODUCT	TRIGGER	DAYS A YEAR	PAID DAYS	WORKING	PRICE A MONTH
Mumbai Basic	≥204.5 mm	1.08	1.08	$1.08 \times 300 = 324 \rightarrow \div 0.6 = 540 \rightarrow \div 12$	₹45
Nagpur Basic	≥47°C	2.20	1.24	$1.24 \times 300 = 372 \rightarrow 620 \rightarrow \div 12$ (2013 alone had 14 hot days; only 5 count)	₹52
Ahmedabad Basic	≥47°C	0.72	0.64	$0.64 \times 300 = 192 \rightarrow 320 \rightarrow \div 12 = ₹27 \rightarrow$ raised to our minimum	₹29

Mumbai Plus	≥64.5 mm	11.88	5.00	Every year has ≥5 heavy days: $5 \times 300 = 1,500 \rightarrow 2,500 \rightarrow \div 12$	₹208
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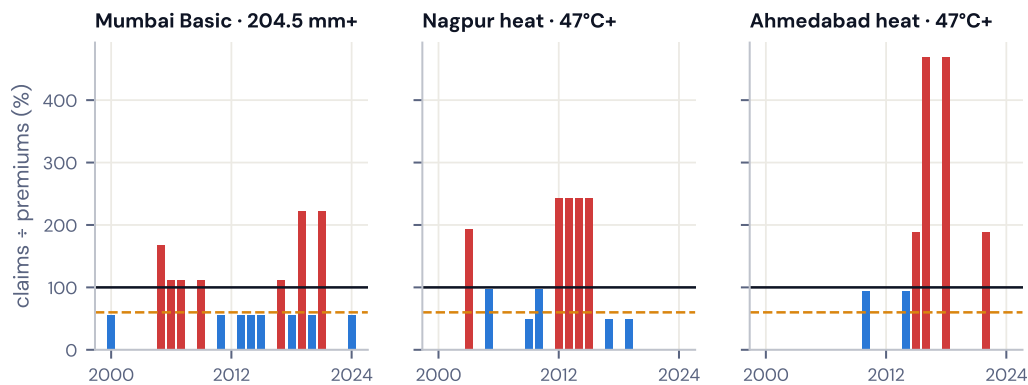
Every year since 2000, Santacruz had at least 5 heavy-rain days (fewest 5 in 2015, most 19 in 2010), so a 5-day cap is reached every year.

WHAT IF...	PRICE	WHY WE DIDN'T CHOOSE IT
Plus with a 3-day cap	₹125	A cheaper option to offer platforms; kept as a variant
Mumbai at "very heavy" (≥115.6 mm, 4.4 days a year)	₹165	A middle option
Nagpur at 45°C (9.7 days a year)	₹168	Too expensive for individuals
Ahmedabad at 45°C (5.4 days a year)	₹100	Too expensive for individuals

6.4 The stress test: would it survive bad years?

We fixed each price from the 25-year average, then replayed every year 2000–2024 as if we had sold the product that year. That year's **loss ratio** = payouts ÷ premiums (100% = break-even).

PRODUCT	WORST YEAR	LOSS RATIO THAT YEAR	YEARS ABOVE 100%	MAX PAYOUT PER PERSON A YEAR
Mumbai Basic	2019 (4 days)	222%	7 of 25	₹1,500
Nagpur Basic	2013 (14 days, 5 paid)	242%	5 of 25	₹1,500
Ahmedabad Basic	2018 (7 days, 5 paid)	469%	4 of 25	₹1,500
Mumbai Plus (5-day cap)	2010 (19 days, 5 paid)	60%	0 of 25	₹1,500



Claims + premiums, year by year, for the three Basic products. Red bars are loss years; solid line 100%, dashed 60% target. Plus is flat at 60% every year.

Meaning: Basic is lumpy (bad years cost 2–5× the premiums), so only a licensed insurer pooling cities and years, with reinsurance, can carry it, never Jio. Plus always hits its cap, so it is perfectly predictable: more like a pre-paid weather allowance than insurance.

6.5 The Mumbai street-by-street check

Same source, Santacruz vs Colaba (about 20 km apart), 2020–2024 monsoons: 89 days when both reported. On 15 of them at least one station had heavy rain (64.5 mm+), and on 5 of those 15 the other station got less than half. Plus the famous day: 26 July 2005, 944 mm vs 73 mm (IMD).

6.6 Business numbers (Q&A only)

NUMBER	WORKING	WHAT IT IS
₹12.5 crore a year	50,000 Mumbai riders × ₹208 × 12	Plus premium pool: the insurer's money, not Jio revenue
₹1.9–3.1 crore a year	15–25% of ₹12.5 crore	Illustrative distribution fees for Jio
₹174 crore a year	50 crore users × 1% × ₹29 × 12	The video's B2C pool (shown as ≈ ₹170 crore); removed because uptake is a guess

Glossary: every term explained

Every word you might hear or need, in one line each.

TELECOM AND THE NETWORK

Microwave link. An invisible radio beam between two towers that carries phone and internet traffic, like a torch beam aimed from one tower to the other. Rain weakens it.

Backhaul. The connections that carry traffic from a tower back into the main network. Done by fibre cables or microwave links.

Fibre (optical fibre). A glass cable, usually underground, carrying data as light. Rain doesn't affect it, so it gives no rain signal.

Tower / site. A place with antennas and radio equipment. Tower companies may own the steel; the operator (Jio) runs the radios and owns the link data.

Frequency (GHz). How fast the radio wave vibrates; 1 GHz = a billion times a second. Higher-frequency links react more strongly to rain.

E-band. Very high-frequency microwave (around 70–80 GHz) used for short, high-capacity links. Very rain-sensitive; can drop out in extreme rain.

Polarization. The direction the radio wave wiggles: vertical (V) or horizontal (H). It slightly changes how rain affects the signal.

Path / path length. The straight line between the two towers of a link, and its length in km.

TSL / RSL. Transmitted signal level (power sent) and received signal level (power arriving), measured in dBm.

dB, dBm. A log scale for power. dBm compares with one milliwatt. 3 dB less = half the power; 10 dB less = one-tenth.

Attenuation / signal loss. How much weaker the signal is when it arrives. Rain adds extra attenuation.

Rain fade. Telecom engineers' name for a link weakening because of rain.

Fade margin / outage. The spare signal a link can lose before it stops working. In extreme rain a link can drop out completely (an outage).

Telemetry. Measurements equipment sends about itself, such as signal levels, every minute or every 15 minutes.

NMS / NOC. Network management system (software that stores telemetry) and network operations centre (the team watching the network).

Network health data. Jio's own records of equipment faults, outages and maintenance. Used to stop a broken link being mistaken for rain.

Operator. A mobile network company: Jio, Airtel, Vi, BSNL.

WEATHER AND MEASUREMENT

Hyperlocal. Weather at the scale of a neighbourhood or street, not a whole city.

Rain rate (mm/h). How hard it's raining: millimetres of water that would pile up in an hour.

Rain gauge. A bucket-like instrument that measures rain at one spot. Very accurate there, but only there.

AWS. Automatic weather station: a gauge plus sensors that reports by itself.

Doppler weather radar. A radar that scans the sky and estimates rain over a wide area. IMD runs about 40 across India.

RADOLAN YW. The German Weather Service's radar rain product (5-minute, ~1 km squares) used as the answer key in our check.

Satellite rainfall. Rain estimated from space. Covers everywhere but is coarse (roughly 10 km squares) and can lag.

IMD. India Meteorological Department, the government weather agency.

BMC. Brihanmumbai Municipal Corporation, Mumbai's city government, which runs its own rain gauges.

Synoptic station. An official weather station reporting at fixed times; the source of our India data.

NOAA GSOD. The US weather agency's public archive of daily station reports worldwide, including IMD's.

Monsoon / north-east monsoon. India's main rainy season is June–September (south-west monsoon). The north-east monsoon brings rain to parts of south India from October to December.

Heatwave. A spell of dangerously high temperature. IMD declares one in the plains when temperatures reach about 45°C.

Path-averaged rainfall. The average rain along a link's whole path. A link can't tell where on the path the rain fell.

Wet antenna. A film of water on the antenna that adds signal loss not caused by rain in the path; corrected (up to 1.5 dB).

Baseline. The signal loss a link would have if it were dry right now. Rain loss = loss – baseline.

Wet/dry detection. Deciding which minutes are rainy by how much the signal wobbles compared with normal.

ITU-R P.838. The international standard formula linking rain rate to signal loss per km: $k = a \times R^b$.

Nowcast / forecast. A nowcast describes rain now or in the next hour; a forecast predicts hours or days ahead. JioMausam observes; it doesn't forecast.

OUR METHOD AND STATISTICS

Dataset. A collection of measurements. Ours: 500 links $\times$ 2 channels $\times$ 15,840 minutes, plus radar.

pycomlink. Free, open-source Python software from KIT, Germany, for turning microwave link data into rain.

Calibration / scale factor (K). One multiplier (1.17) that matches our totals to radar's on the tuning days.

Tuning days / unseen test days. 10–12 May used to fit K; 13–20 May used only for scoring. Prevents marking our own homework.

Correlation (r). How closely two series rise and fall together: 1 = perfectly, 0 = not at all. Doesn't check amounts.

Bias. Our total $\div$ the reference total. 0.82 means we read 18% low.

IDW (inverse distance weighting). Filling in a map: each square averages nearby links, nearer ones counting more (weight $1 \div \text{distance}^2$).

Zone / square / grid. The map is split into squares (about 3 $\times$ 4 km in our check). A grid is the full set of squares.

Trigger / event. A rule that turns rain into a yes/no decision. Our event: 10 mm+ in any 3 hours in a zone on a day.

Events caught (hit rate). Of all real events (radar said yes), the share we also flagged: 68%.

Unconfirmed trigger. We flagged an event radar didn't see: 18% of our triggers. Not called "false" because radar can miss too.

Agreement. Share of all decisions where both said the same: 94.6%, inflated by dry days.

Threshold. The cut-off level, e.g. 10 mm. We also tested 5 and 15 mm.

Fusion. Combining several sources (links, radar, gauges, satellite) into one better estimate.

Held-out gauges. Gauges hidden from the system while it's tuned, used only for scoring.

Back-test. Running the system on past (archived) data to see how it would have done.

Accuracy gates. Our pass marks: $\geq 80\%$ of trigger events caught, totals within $\pm 10\%$, under 15% unconfirmed.

BUSINESS AND STRATEGY

B2B / B2C / B2B2C. Selling to businesses / to consumers / to businesses who pass it on to consumers.

Proving ground. Where you test something on yourself first (Reliance) before selling it.

Decision infrastructure. Not a map: alerts and triggers that plug into a business's own decisions.

API. A way for one company's software to request data from another's automatically.

Quick commerce. 10–30-minute grocery and goods delivery (e.g. Zepto, Blinkit, Instamart).

Moat. What makes a business hard to copy. Ours: integration, not the sensor.

Integration. Connecting layers: Sense (network) $\rightarrow$ Understand (JioMausam) $\rightarrow$ Reach (Reliance) $\rightarrow$ Protect (insurers).

KPI. Key performance indicator: a number you track to judge success, e.g. site visits avoided.

Matched zones. Comparing zones using JioMausam with similar zones not using it, on the same days.

Pilot / gates / stop-or-scale. A small test with pass/fail points; pass all three to grow, otherwise stop.

JFS. Jio Financial Services: a separately listed Reliance group company in financial services.

INSURANCE

Parametric insurance. Pays a fixed amount when a measured trigger is crossed (e.g. rain above a level), without assessing each person's loss.

Premium. What the customer pays for cover.

Payout / benefit. What the insurer pays when the trigger happens (prototype: ₹300).

Loss ratio. Payouts $\div$ premiums. We assume 60%: ₹60 of every ₹100 goes back to customers.

Cap. The most paid days in a year (5).

Actuary / actuarial validation. An insurance mathematician who sets prices and checks they're safe.

Underwrite. To take on the insurance risk and pay claims. Only a licensed insurer can.

Reinsurance. Insurance for insurers, protecting them in very bad years.

Basis risk. When the trigger and real life disagree: rained on but no payout, or the reverse.

Moral hazard / gaming. People changing behaviour to collect a payout (e.g. logging in only when it rains).

Adverse selection. People buying cover only when they expect bad weather.

IRDAI. Insurance Regulatory and Development Authority of India, the insurance regulator.

Regulatory sandbox. A regulator's scheme letting firms test new products in a limited, supervised way.

Shadow mode. Running the system and recording would-be payouts without paying any money.

WBCIS. India's government weather-based crop insurance scheme: an existing parametric product.

Question bank

Answer in two or three sentences, then stop. End by pointing back to the strategy: combine sources, prove it inside Reliance, gates before scale.

The 12 questions to know cold

Q What is JioMausam, in one line?

A weather-intelligence layer built from signals Jio's microwave network may already carry, combined with radar and gauges and turned into zone-level alerts that businesses act on. Insurance is one optional application on top.

Q Are you claiming Jio already has a working weather network?

No. We know the physics works, and our own links-alone test shows it isn't enough on its own. We don't yet know Jio's usable coverage, Indian performance or commercial value. So we're asking for 90 days to measure those, with explicit gates and a stop-or-scale decision.

Q How did you get 68%?

A public dataset from KIT in Germany: 500 real phone-network links, signal every minute for 11 days, with German weather radar as the answer key. Rain weakens the signal; a standard formula turns that loss into rain. For every ~4 km zone each test day we asked: 10 mm in 3 hours? Radar said yes 1,639 times; links also said yes on 1,120. $1,120 \div 1,639 = 68\%$. Below our 80% bar, which is why we never use links alone.

Q Why 80%?

Our go/no-go bar for the pilot, set deliberately above the 68% links alone achieved, so passing proves the combined system adds value. It sits with two other gates so you can't pass by triggering every day, and an insurer or use case can tighten it.

Q Why would Swiggy or Zomato pay?

We don't assume they will; that's why we prove value inside Reliance first. The value case: rain shifts demand and rider supply zone by zone, so better zone decisions mean fewer late orders, less wasted incentive spend and fewer cancellations. Gate 3 measures it with matched zones before we sell outside.

Q Why can't Google or a weather company do this?

They forecast and they sell weather data, so they're potential inputs or partners. What they don't have is the signal inside Jio's network, or Reliance operations to prove it on. Our edge is integration, not the sensor.

Q What stops Airtel from copying you?

Nothing in the physics; the method is published. The edge is speed and integration: Jio's network data plus Reliance's operations, reach and financial ecosystem. If Airtel does it too, that validates the market, and businesses will buy whichever is more accurate in their zones.

Q What if Jio doesn't have enough microwave links?

Then Gate 1 tells us in the first month, cheaply. The system still runs on radar, gauges and network-health data for operations, and we can add gauges on Jio sites where links are thin. The question becomes how much Jio's links add, not whether the business exists.

Q Why include insurance at all?

It's where we started and it matters to workers. But it launches only after the accuracy and actuarial gates pass, with a licensed insurer carrying the risk. If it never launches, the intelligence business still stands.

Q How did you price Kavach?

From 25 years of IMD station records we counted how often each trigger happens, capped paid days at 5 a year, and applied: $\text{paid days} \times ₹300 \div 60\% \div 12$. ₹300 and 60% are prototype assumptions; an actuary sets final prices.

Q What do you need from Jio for the pilot?

Mostly access, not money: link telemetry and network-health logs for one city, a small data team with a meteorology partner, IMD/BMC data, and one Reliance operation willing to run a shadow test. We'd cost it properly with Jio once Gate 1 shows what data exists.

Q What is your single biggest risk?

Coverage: whether Jio has enough usable microwave links where it matters. That's exactly why it is Gate 1, and it costs almost nothing to check.

Quick-fire: what, who, why, how, when, where

What is JioMausam?

A weather-intelligence layer: signals Jio's microwave links may carry, combined with radar and gauges, turned into zone-level alerts and triggers.

Who is it for first?

Jio's own network operations, then JioMart and MyJio; then delivery platforms, logistics, cities and insurers.

Why Jio?

It may already own the sensor (microwave links) and can connect sensing, intelligence, reach and protection.

Why not insurance first?

Who pays is unproven and links alone aren't accurate enough to pay on. Insurance comes after the gates.

How accurate is it?

Links alone caught 68% of rain events in our check; our bar for the combined system is 80%, with totals within $\pm 10\%$ and under 15% unconfirmed.

How much does the pilot cost?

Mostly access and a small team on existing data; costed with Jio once Gate 1 shows what data exists.

When would it launch?

Reliance live and first clients in months 4–12 if the gates pass; cities and enterprises in Year 2; a national API in Year 3.

What could kill it?

Too few usable links, no mix clearing the gates, or no measurable value inside Reliance. Each is a cheap stop.

What is Mausam Kavach?

An optional parametric insurance application: a fixed payout when a weather trigger is crossed, underwritten by a licensed insurer.

Who pays?

Inside Reliance: savings. Outside: businesses pay for alerts and data. For Kavach, who pays is unproven and tested in the pilot.

Why now?

Zone-level delivery decisions matter more than ever, extreme rain is rising, and operators keep adding microwave backhaul.

How does it work?

Rain weakens the beam → we measure the extra loss → formula turns it into rain → combine with radar and gauges → zone alerts.

How do you make money?

Intelligence fees (alerts, API, data), savings inside Reliance, and optional distribution fees if Kavach launches.

How long until we know?

Ninety days: coverage (days 1–30), accuracy (15–60), value (61–90).

Where first?

Wherever the coverage audit finds enough usable links; back-test on archived Mumbai monsoon data.

What do you need from us?

Link telemetry and network-health logs for one city, one Reliance operation for a shadow test, and 90 days.

Questions about our video

Judges saw the video. Own every change; never pretend it didn't say something.

Q Your video said every phone is a thermometer. Why isn't that in your deck?

It's real research (Overeem et al., 2013), but it reads phone battery temperatures through an app, which needs users' consent; it gives city-wide daily averages, not street-level readings; and phones heat up from use and charging. Not reliable enough to pay money on, so heat now comes from IMD stations and satellite.

Q Your video said the network measures 46.2°C in Nagpur. Can it?

No. That screen was illustrative. Microwave links sense rain, not temperature. Heat data comes from IMD stations and satellite through the same engine.

Q Your video said ₹29 a month for heat and rain. Now prices are ₹45, ₹52, ₹208. Why?

When we priced it with 25 years of IMD records, risk differed a lot by city. A flat ₹29 would lose money in Mumbai and Nagpur. ₹29 survives as our minimum price, which is what Ahmedabad gets.

Q Your video said ₹170 crore a year. Where did that go?

It was 1% of Jio's users × ₹29 × 12: the insurer's premium pool on a guessed uptake, not Jio revenue. We removed it because we can't back the uptake.

Q Your video said the payout lands in JioPay within two hours.

A licensed insurer would pay through its own rail. ₹300 and the speed are prototype targets we'd measure in shadow mode.

Q Your video said JFS partners would underwrite it.

We have no insurer partner. JFS is a separately listed group company and a natural partner, but any deal would be at arm's length, and only after the accuracy and actuarial gates.

Q Your video said "Why only Reliance?". Now you say integration. Isn't that weaker?

It's more honest. Others could do parts of this; Reliance can connect all the layers and prove it on its own operations. "Only" was easy to attack.

Q Your video said "zero new hardware". Now there are new gauges.

Mostly existing infrastructure. A few gauges on Jio sites only where links are thin; that's our safety net, not the plan.

Q Your video said the pilot was rain in Vidarbha and heat in Ahmedabad.

A 90-day pilot starting now would miss Vidarbha's monsoon (June–September) and Ahmedabad's heat season (April–June). So we back-test archived Mumbai monsoon data, let the coverage audit choose the live city, and test heat in months 4–12.

Q Your video said New Energy could use it for solar output. Why drop that?

Solar output depends on sunshine and cloud, which rain links don't measure. We kept only uses our data can actually serve.

Q Your video said Jio would use it for rain-fade prediction.

We now say detection, not prediction: can we tell rain-related signal loss from equipment faults before sending a crew? That's something we can test.

Q Your video said heat could cost India 34 million jobs.

The precise ILO estimate is working hours equal to about 34 million full-time jobs by 2030. Heat is no longer our lead, so it isn't in the deck.

Q Your video said lakhs of stations in every district.

We learned only microwave links count, and nobody outside Jio knows how many usable links there are. That's Gate 1.

Q Your video was about farmers, riders, workers and vendors. Now it's about businesses.

Workers still benefit through Kavach later. Businesses that decide many times a day, starting with delivery, are a clearer first market.

Q Why should we trust you when your pitch changed this much?

Because it changed for evidence. We tested our own idea, found where it broke, and kept what survived. That's what the 90-day gates are for too.

Competition and alternatives

Judges will test whether someone bigger already does this. Never attack competitors; position them as inputs, partners or proof the market is real.

Q Google and DeepMind already do AI weather forecasting. Why you?

Forecasting and observing are different jobs. AI forecasting models predict weather hours to days ahead from global data. JioMausam observes rain happening now, zone by zone, and turns it into operational alerts. They complement each other: a forecast says rain is likely this afternoon; JioMausam says it is raining in Zone A now. A forecast model could even use our observations as an input.

Q Weather companies already sell hyperlocal weather APIs (e.g. The Weather Company, Tomorrow.io, Skymet).

Yes, and platforms already buy from them. Their hyperlocal products are mostly models built on radar, satellite and stations. None of them has Jio's network signal. If our links add accuracy, that's a real edge; if they don't, our pilot's "radar + gauges, no links" baseline shows it and we won't pretend otherwise. Some of these firms could be partners or data buyers.

Q Hasn't this been tried commercially already?

The idea isn't new, which is what makes it credible. Researchers proved it in Israel (2006) and mapped a whole country in the Netherlands (2013), and weather startups and telecom vendors have built or piloted products on wireless-signal data. What's new is doing it in India, on Jio's network, combined with IMD data and proven inside Reliance operations.

If pushed: Researchers and companies have worked on this for over a decade. We'd rather show our own check than quote other people's claims.

Q IMD has radars and is expanding under Mission Mausam. Doesn't that make you redundant?

No. IMD is an input and a natural partner. Its radars and gauges are in our mix. IMD's job is public forecasts and warnings; ours is turning observations into decisions for specific operations. More IMD radars make JioMausam better, not obsolete.

Q Mumbai already has BMC rain gauges and hyperlocal weather apps.

Exactly, and they're inputs we combine, not rivals. The real question is whether Jio's links add information between gauges. The pilot measures that directly with the "radar + gauges, no links" run.

Q What stops Airtel, Vi or BSNL?

Nothing technical; the method is open. The edge is integration and speed: Jio's network signal plus Reliance's operations to prove value on, its enterprise reach and its financial ecosystem. A competitor doing it would confirm the market.

If pushed: "Could you work with other operators?" Possibly later: more links mean better coverage. We'd start with Jio's own network.

Q Why not tower companies? They own the towers.

They own the steel and the sites. The radio links and their signal data belong to the operator running them. That's Jio.

Q Delivery platforms can build this in-house.

They can and should build their own decision logic. What they can't build is the raw network signal. Our product is the intelligence feed they plug into their own dispatch systems.

Q Why not just buy weather data and skip Jio's links?

That is our pilot baseline. If links add nothing, we say so; then the case rests on the decision layer and Reliance's distribution, and the moat is weaker. Better to learn that in 90 days than after a big launch.

Q Insurers already sell parametric weather insurance (e.g. crop insurance).

Yes: India already runs a weather-index crop insurance scheme (WBCIS), so parametric cover is familiar to insurers and the regulator. That's why Kavach plugs into licensed insurers instead of competing with them. We supply trigger data.

Q Satellites cover all of India. Why not use them?

Satellite rainfall is open and covers everything, but it's coarse and can lag, so it's our fallback, not the main source. The pilot evaluates satellite sources on coverage, latency, resolution and licensing.

If pushed: Open satellite rainfall products are roughly 10 km squares, far coarser than zone decisions need.

Q Why not put IoT rain sensors everywhere?

Hardware at scale means capex, maintenance and theft. We add gauges only where links are thin, on Jio sites that already have power, security and connectivity.

Q Why not crowdsource from smartphones?

Phones don't measure rain; user reports are noisy and sparse in heavy rain. It could become a confirmation signal later, but it isn't the core.

Q Why not use 5G or phone-to-tower signals?

Backhaul links are fixed point-to-point paths with known length and frequency, so the physics is clean. Phone-to-tower signals vary with movement, buildings and indoor use, so rain is much harder to separate out.

The problem and the idea

Show the problem is real without overclaiming.

Q Is this a real problem? Platforms already handle rain.

They handle it with city-level information and manual calls. Platforms charging a rain fee shows rain clearly matters to their economics. Our claim is narrow: zone-level precision could improve those decisions, and Gate 3 measures whether it does.

Q One day in 2005 isn't evidence.

Agreed, so we checked recent years too. In the 2020–2024 monsoons, Santacruz and Colaba both reported on 89 days. On 15 of them at least one station had heavy rain (64.5 mm+), and on 5 of those 15 the other station got less than half. Rain differing sharply across one city is routine, not a one-off.

Q Isn't this a solution looking for a problem?

We started from a problem, riders losing income and businesses deciding blind in rain, and stress-tested our way to the solution. And we test value before scaling, so if the problem isn't worth solving, Gate 3 says so.

Q Why rain and not floods or air quality?

Rain is what microwave physics measures. Flood alerts for sites are a downstream use of rain data. Air quality can't be measured this way.

Q Why Mumbai?

Most heavy-rain days of our three cities (about 12 a year), dense operations, and the clearest example of rain varying within a city. But the live-test city is chosen by the coverage audit, not by us.

Q "Decision infrastructure" sounds like jargon.

It means we don't sell a weather map. We sell alerts and triggers that plug into a business's own decisions: dispatch, pausing routes, incentives, payouts.

The technology

Plain answers; admit what the pilot must measure.

Q How exactly does rain weaken the signal?

Raindrops absorb and scatter microwave energy. The heavier the rain, the bigger the loss. The relationship is an international standard (ITU-R P.838) and depends on the link's frequency, polarization and length.

Q Which frequencies work?

Links in our test ran from about 6 to 39 GHz; higher frequencies react more strongly to rain. Very short, very high-frequency links are highly sensitive but can drop out in extreme rain. The coverage audit lists which of Jio's bands are usable.

Q Jio is mostly fibre. Doesn't that kill the idea?

In dense cities fibre may dominate; fibre gives no rain signal. That's why coverage is Gate 1. Where links are thin we lean on radar, gauges and new gauges on Jio sites. The business doesn't depend on Jio's links alone; the pilot measures what they add.

Q In very heavy rain, doesn't the link just drop?

Some links can, at extreme rain. A drop-out itself tells you rain passed a high level, and radar and gauges cover the gap. We treat outages as information, not missing data.

Q How do you tell rain from an equipment fault?

Three ways: the pattern (rain builds and fades; faults are sudden), neighbours (rain hits nearby links together; a fault hits one), and Jio's network-health flags for faults, outages and maintenance. That's why network health is a source on slide 4.

Q What is the wet-antenna problem?

Water on the antenna adds loss that isn't rain along the path. We correct it with a standard method (Schleiss et al., 2013), allowing up to 1.5 dB.

Q What about fog, humidity and temperature?

Their effect is much smaller than rain at these frequencies and changes slowly, so the dry-weather baseline absorbs it.

Q Does Jio even store this data? How often?

Operators' network management systems record signal levels; often as 15-minute minimum and maximum values. The Netherlands country-wide maps were built from exactly that kind of operator data. How often and how far back Jio's data goes is part of Gate 1. Live minute-level data would need polling.

Q A link averages rain along its path. How do you know where it rained?

Right: one link gives an average. Many crossing links plus radar and gauges locate the rain; that's the fusion step that builds the grid.

Q What resolution will you get?

It depends on link density and layout. Our German check used ~4 km zones. We don't promise 1 km everywhere; the pilot finds the real resolution city by city.

Q How fast are alerts?

Near-real-time; how near depends on how fast telemetry can be pulled. Operational decisions need minutes, not seconds. The pilot measures it.

Q Can JioMausam forecast?

It observes current rain. Short-term forecasts could be built on top later by tracking movement, but we don't claim that.

Q Is the ITU formula valid for Indian monsoon rain?

The formula assumes typical raindrop sizes; monsoon rain can differ. That's exactly why local calibration against Indian gauges is part of Gate 2.

Q How much data and computing does this need?

Modest: a few numbers per link per minute. Jio already processes far larger telemetry every day.

Q Will Jio's security team allow this?

Data stays inside Jio. JioMausam only outputs aggregated rain per zone, never network layout or customer data. Network performance readings aren't personal data.

Q How much does a raindrop actually weaken the signal?

In our example, heavy rain (about 25 mm/h) added about 24 dB of loss over 8.5 km at 23 GHz: the received power dropped to well under 1% of normal. That big a change is easy to detect.

Q Why average the two channels?

Each link has two directions; averaging them reduces random noise.

Q Why a 60-minute window for wet/dry?

Long enough to see the signal wobble that rain causes, short enough to react within the hour. It's the standard approach in the toolbox.

Q Why $1.12 \times$ the 80th percentile?

Each channel learns its own normal jitter; the 80th percentile is a "usually dry" level and 1.12 adds a small safety margin. It's a tuning choice from the method, not a magic number.

Q What if a link's power is changed automatically?

Some radios raise their transmit power in rain. That's why we use sent minus received, not received power alone.

Q Why 4 km zones and not 1 km?

We merged radar's 1 km grid 4×4 to match how far apart links typically are. Finer zones would mostly be guessed, not measured.

Q Why place each link at its midpoint?

It's the simplest choice for a path average. Better methods spread the value along the path; ours is a floor.

Our 68% check

Know this cold. Answer, then return to strategy.

Q Why German data and not Indian or Jio data?

Jio's data isn't public. This is the standard public research dataset with real links and a radar answer key. Jio's real number is exactly what the pilot measures.

Q Only 11 days?

Yes: it's a check of the method, not a final validation. It still gives 14,256 yes/no decisions on 8 unseen days. The pilot uses a full monsoon.

Q May in Germany isn't a monsoon.

Agreed. Monsoon rain is heavier and longer; heavier rain gives a stronger signal but more drop-out risk. That's why Gate 2 back-tests archived Mumbai monsoon data.

Q Radar isn't ground truth.

Agreed; it was the reference in that dataset. That's why we say "not confirmed by radar", not "false". The pilot checks against held-out rain gauges.

Q Did you tune it to look good?

We tuned one scale factor on the first 3 days only, then tested on the 8 days it had never seen.

Q One scale factor for 500 links sounds crude.

It is, deliberately. Production systems calibrate locally. Our result is a floor, not a ceiling.

Q How closely did links track radar?

Network-wide hourly match 0.95; a typical single link 0.88; map level 0.78. The trigger test is harsher: a 9 mm estimate against an 11 mm event counts as a miss.

Q Why do links read 18% low?

The tuning days had very little rain, so the scale factor was fitted on light rain, and path-averaging smooths peaks. Local calibration in the pilot addresses it.

Q Why 10 mm in 3 hours?

Our heavy-shower rule for the check. We also tried 5 mm and 15 mm: 64% and 54% caught. None got near 80%, so the conclusion doesn't depend on the rule.

Q Which software?

Python with pycomlink, the open-source toolbox from KIT that implements the published method, plus our own code for the zone map and the comparison.

Q Can we check your work?

Yes. The data is public, the code is in our repository, the workbook recomputes 68% from the 14,256 decisions, and our Test Lab reruns everything in a browser.

Q Did you build a model?

No new model and no machine learning. We ran the published method on public data and calculated the result ourselves.

Q Is 68% good or bad?

For one source with simple processing it's a reasonable floor. For paying claims it isn't enough. That gap is why the product combines sources.

Q What is the 18% unconfirmed?

Of our 1,364 triggers, 244 weren't seen by radar. Some are our errors, some may be radar misses.

Accuracy gates and combining sources

Q Why $\pm 10\%$ on rain totals?

If amounts are off, prices and payouts are off. Our links-alone check read 18% low, so this gate forces calibration.

Q Why under 15% unconfirmed?

So a system can't "pass" the 80% gate by triggering every day. Our check had 18%.

Q Who sets the gates?

We set them as the pilot's go/no-go. Jio, and for Kavach an insurer, can tighten them.

Q What are held-out gauges?

Gauges kept hidden from the system while it's tuned, and used only to score it, so we're not marking our own homework.

Q What if no combination clears 80%?

Then we don't scale externally and Kavach stays in shadow or is dropped. We'd keep only uses that need less precision, like network operations, if Gate 3 shows value.

Q How do you know what each source costs?

We don't claim prices. The slide shows who provides each source. The pilot puts a rupee figure on each, and the rule picks the cheapest mix that clears the bar.

Q What is "network health" data?

Jio's own alarms and logs: equipment faults, outages, maintenance windows. It removes signal drops that aren't rain.

Q What would new gauges on Jio sites cost?

Priced in the pilot. They're only for areas where links are thin, and Jio sites already have power, security and connectivity.

Q When would you buy commercial data?

Only if it lifts accuracy enough to justify its cost. Otherwise not.

Reliance as the proving ground

Q Why start inside Reliance?

Jio owns the data and the decisions, savings are measurable, and there's no sales cycle. It proves value before anyone outside has to pay.

Q Doesn't Jio's network team already know rain fade?

It may. Then Gate 1 confirms cheaply what weather information is available. If it doesn't, that's new value. Either way the same telemetry feeds the weather layer.

Q How would network operations save money?

By not sending crews for rain-related signal loss that recovers on its own, prioritising real faults, and warning sites at flood risk. Gate 3 measures site visits avoided, fault-resolution time and weather outages correctly identified.

Q What does JioMart get?

Rain-aware delivery promises, dispatch and zone pauses. Measured by late deliveries, cancellations and rider availability.

Q You said it isn't a consumer app, but MyJio sends alerts.

MyJio tests reach inside an existing app. We're not building a standalone consumer weather app business.

Q Will Reliance businesses actually cooperate?

That's part of the ask: one operation for one shadow test. The coverage audit decides which city.

Market, customers and selling

Q What exactly do you sell?

Zone alerts (rain starting, intensity), live API feeds, historical rain records for audits and claims, and trigger data for insurers.

Q How will you price the intelligence product?

After the pilot, benchmarked to the value Gate 3 measures. Likely a subscription per city or zone, or by API volume. We won't guess numbers before we have evidence.

Q What is the market size?

We deliberately don't headline one; we'd build it bottom-up after the pilot. For Kavach alone, 50,000 Mumbai riders on Plus is about a ₹12.5 crore premium pool a year, which is the insurer's money, not Jio revenue.

If pushed: If asked about the video's ₹174 crore: 50 crore users × 1% × ₹29 × 12. It is an illustrative premium pool, B2C uptake is unproven, so we removed it.

Q Why delivery first?

It makes the most rain-sensitive real-time decisions, zone by zone, many times a day.

Q Why would BMC or a city buy?

Flood response, pump deployment, traffic management. Public-sector sales are slower, so cities are tier 2.

Q What about agriculture?

Rainfall records for crop insurance and advisories; rural areas may rely more on microwave backhaul. Tier 3, and unproven.

Q How do you reach customers?

Potentially through Jio's enterprise sales relationships and Reliance's partners. That's the "Reach" layer.

Q Why buy from Jio rather than a weather firm?

Only if Jio's links add accuracy in the buyer's zones, plus an auditable multi-source trigger and integration with Reliance's ecosystem. If links don't add accuracy, we don't claim an edge.

Why Jio, the moat and the economics

Q The method is open-source. Where is the moat?

Access to Jio's private link data, calibration history built over seasons, integration with Reliance operations, and distribution. The method is open; the signal isn't.

Q Why should Reliance invest?

Low capex because the infrastructure exists, internal savings first, optional upside, and a strategic data asset. And it costs 90 days to find out.

Q Where are your revenue projections?

None on the slide on purpose. We'd rather show value measured in Gate 3 than invented forecasts.

Q Savings inside Reliance aren't revenue.

Correct, which is why slide 7 separates external revenue from internal value.

Q JFS isn't part of Reliance any more.

It's a separately listed group company and a natural partner. Any arrangement would be at arm's length, with a licensed insurer carrying the risk.

Q What does Jio actually risk?

A small pilot. No insurance risk; shadow mode protects reputation.

Q How big are distribution fees?

Illustratively, 15–25% of a ₹12.5 crore premium pool is ₹1.9–3.1 crore. Small, which is exactly why insurance is not the business.

Kavach (insurance)

Keep Kavach clearly optional and clearly the insurer's risk.

Q What is parametric insurance?

It pays a fixed amount when a measured trigger is crossed, like rain above a level in your zone, without assessing individual losses.

Q Why can't Jio be the insurer?

Insurance needs an IRDAI licence and capital. Jio supplies data; a licensed insurer underwrites and pays.

Q What is basis risk?

A worker gets rained on but no payout triggers, or the reverse. Small zones, multi-source triggers and a review process reduce it, which is why the accuracy gates come first.

Q Won't people log in just to collect payouts?

Eligibility requires being already active in the zone before the event began, checked against platform activity logs.

Q Won't people buy only when heavy rain is forecast?

That's adverse selection. The answer is season-long or annual cover, or platform group cover, with no buying in mid-event.

Q Who pays: platforms or riders?

Unproven: platform-sponsored, co-funded or individual. The pilot tests it.

Q Why would a platform pay ₹2,500 a year for at most ₹1,500 of payouts?

The gap is the insurer's costs at an assumed 60% loss ratio. A platform could fund the allowance itself using our trigger data, and Jio earns from the data either way.

Q How is the money paid? JioPay?

Through the insurer's payment rail (UPI or bank). We don't depend on any particular wallet. Shadow mode measures the speed.

Q What is the IRDAI sandbox and shadow mode?

Shadow mode computes would-be payouts without moving money. The regulator's sandbox lets insurers test new products within limits. Real payouts come only after the gates.

Q Why ₹300?

A round prototype figure, roughly part of a day's earnings. The insurer or platform would set the real benefit.

Pricing

Every price is illustrative. Know the formula and one worked example.

Q Walk me through ₹45.

Mumbai sees ≥ 204.5 mm in a day about 1.08 times a year. $1.08 \times ₹300 = ₹324$ of expected payouts; $\div 0.6 = ₹540$ a year; $\div 12 = ₹45$ a month.

Q And ₹52, ₹29, ₹208?

Nagpur: 2.2 days a year at $47^\circ\text{C}+$, but capped at 5 paid days a year that averages 1.24 paid days $\rightarrow ₹52$.

Ahmedabad: 0.64 paid days $\rightarrow ₹27$, raised to our ₹29 minimum price. Mumbai Plus: ≥ 64.5 mm happens about 12 days a year, every year at least 5, so it always pays 5: $5 \times 300 \div 0.6 \div 12 = ₹208$.

Q Why a 60% loss ratio?

An illustrative assumption: ₹60 of every ₹100 goes back as payouts, ₹40 covers costs and margin. An actuary sets the real figure.

Q Why cap at 5 days a year?

It limits the worst-year cost and keeps the price predictable. With a 3-day cap, Plus would be ₹125.

Q Why ₹29 for Ahmedabad when the maths says ₹27?

Our chosen minimum price, so even very low-risk cover pays for basic administration.

Q Why 47°C and not 45°C ?

45°C happens about 10 days a year in Nagpur, which would price Basic at about ₹168 a month. 47°C targets rare extremes at an affordable ₹52.

Q Why ≥ 204.5 mm for Mumbai Basic?

That's IMD's "extremely heavy rain" level: rare (about once a year), so Basic stays cheap. IMD's "heavy rain" level (64.5 mm) is the broader Plus product. A very-heavy trigger (115.6 mm) would price at ₹165.

Q Your live trigger is 10 mm in 3 hours. Why price on 64.5 mm a day?

₹208 isn't priced from that prototype rule. It uses 25 years of daily IMD heavy-rain history, capped at five paid days. The pilot measures the 3-hour trigger's real frequency and recalibrates before any launch.

Q What happens in bad years?

We replayed every year from 2000 to 2024. Worst years: Mumbai Basic 2019 paid 222% of premiums, Nagpur 2013 242%, Ahmedabad 2018 469%. Plus was 60% every year because it always hits its cap. Bad years are why a licensed insurer pools risk across cities and years, with reinsurance.

Q Is NOAA's data reliable?

It's NOAA's public archive of IMD's own station reports. We kept only years with at least 330 days of reports, including 110 in June–September; all 25 years passed. Missing days count as no trigger, so if anything we slightly undercount.

Q Why these three cities?

Mumbai is India's rain extreme; Nagpur and Ahmedabad are heat extremes. Together they show one flat price can't fit India.

Q Climate change makes 25-year averages wrong.

Trends matter; studies report a sharp rise in widespread extreme rain over central India since 1950. An actuary would weight recent years and add a loading. Our numbers are a starting point.

Q Would a rider pay ₹45 a month for about one payout a year?

Maybe not, which is why Basic is optional and who pays is tested. Basic is cheap cover for rare extremes, not an income product.

Pilot and roadmap

Q Is 90 days realistic?

Yes, because nothing new is built first. The audit is desk work on existing telemetry, the back-test uses archived data, and the live test is about a month. The phases overlap.

Q Which city, and when?

The coverage audit chooses. A live test now would run in a north-east-monsoon city (rain from October to December); Mumbai's 2027 monsoon falls in months 4–12.

Q What if it doesn't rain in days 61–90?

The back-test replays archived monsoon events, including network-operations value. The full live test runs in Mumbai's 2027 monsoon.

Q What are matched zones?

Zones using JioMausam compared with similar zones that aren't, on the same days. The difference is the value.

Q What are your stop criteria?

Any gate fails: stop, or narrow to what passed. No external scaling until all three pass.

Q Year 2 says "Basic at recharge". Didn't you say B2C is hard?

It's optional and only happens if Kavach passes its gates and an insurer partners. Recharge is just a distribution channel.

Q How do you get to a national API by Year 3?

City by city: audit coverage, calibrate locally, launch where coverage supports it.

Feasibility and scalability

Q Is it technically feasible?

The physics is proven, the tools are open, the data exists in operator systems, and we ran the method ourselves on real link data.

Q Operationally?

A small team on existing telemetry; no new hardware except optional gauges where links are thin.

Q Financially?

Low upfront cost, internal savings first, and a 90-day decision point before any big spend.

Q Does it scale?

The software scales with the network. Each new city needs a coverage audit and local calibration, and different climates need their own calibration.

Q What skills does it need?

Data engineering, signal processing and a meteorology partner, for example IMD or an academic institute.

Q Networks change. Links get replaced by fibre.

Yes, coverage evolves, so we re-audit. Because we combine sources, quality degrades gracefully instead of failing.

Curveballs

Q Why now?

Quick commerce makes zone-level rain decisions worth more; extreme rain events are rising; operators keep adding microwave backhaul; and IMD keeps adding radars, which are inputs we combine.

Q If Reliance says no, what happens?

Then the 90-day audit is the cheapest way to say no with evidence instead of opinion.

Q If you had to cut one thing, what?

Kavach. The intelligence business stands without it.

Q What did you learn doing this?

That a flattering number (0.95) can hide the number that matters (68%), and that testing your own idea is the fastest way to improve it.

Q Explain it to a 10-year-old.

When it rains, the invisible radio beams between phone towers get weaker. We measure how much weaker to work out where it's raining, and tell businesses which areas to act on.

Q How is this "Different by Design"?

It uses infrastructure Jio already runs as a sensor, it's designed around decisions instead of maps, and it's built to be tested and stopped cheaply instead of launched on faith.

Backup plans: if an assumption fails

Every main assumption has a test, a Plan B and a Plan C. Because we combine sources, JioMausam gets weaker gracefully instead of failing outright, and every gate is a cheap place to stop.

ASSUMPTION	HOW WE FIND OUT	IF IT FAILS: PLAN B	PLAN C
Jio has enough usable microwave links where it matters	Gate 1 coverage audit (days 1–30): link counts, frequencies, path lengths, telemetry	Run the layer on radar + gauges + satellite, plus network-health data for Jio's own operations; focus first on areas where microwave is dense	Add rain gauges on Jio sites where links are thin; if coverage is poor everywhere, the Jio-specific edge is small, so stop external scaling
Jio's telemetry is stored often enough and far enough back	Gate 1: sampling interval, history kept, access route	Use 15-minute minimum/maximum data (it worked country-wide in the Netherlands); start live polling for forward collection	Use link alarm and outage logs as a coarse rain signal; back-test on whatever history exists
Combining sources reaches $\geq 80\%$ of trigger events, $\pm 10\%$ totals, $< 15\%$ unconfirmed	Gate 2 back-test on archived Mumbai monsoon data against held-out gauges	Local calibration, more gauges, better fusion; use the system for decisions that need less precision (network ops, soft alerts)	Kavach stays in shadow or is dropped; no external scaling
Jio's links add accuracy beyond radar + gauges	Gate 2: "radar + gauges, no links" baseline vs every mix with links	Keep links only where they add value (gaps between gauges, rural backhaul); sell the decision layer on the best mix	If links add nothing anywhere, say so: the moat shrinks to integration and distribution, and the go/no-go is decided on that basis
A Reliance operation sees measurable value	Gate 3: matched zones with vs without JioMausam (site visits, fault time, delays, cancellations, incentive spend)	Try another internal operation or a different decision (e.g. flood alerts for sites, staffing)	Stop. A cheap, evidence-based no
External businesses will pay	After Gate 3: paid pilots with one platform and one logistics firm	Bundle alerts into Jio's enterprise connectivity offers as a value-add	Keep it internal, or license data to insurers and agriculture
A licensed insurer partners for Kavach	Months 4–12: shadow results shared with insurers; regulator's sandbox	Platforms self-fund a weather allowance using our trigger data (no insurance needed)	Drop Kavach; the intelligence business is untouched
The price assumptions hold (₹300, 60%, 5-day cap)	Actuarial review; shadow-mode payout data	Change the benefit, cap or trigger (e.g. a 3-day cap brings Plus to ₹125)	Platform group cover instead of individual sales
Who pays: platforms, co-funding or individuals	Pilot willingness-to-pay tests with platforms and riders	Co-funded model (platform + rider)	Data-only product for platforms' own schemes
Security and legal teams allow the data use	Early review with Jio's security and legal teams	Process everything inside Jio; release only aggregated rain per zone	Use the layer internally only (network ops)
Accuracy holds in extreme rain (links can drop out)	Gate 2 back-test on the heaviest archived events	Treat drop-outs as "rain above a level"; lean on radar and gauges in those minutes	Flag extreme events for manual review before any payout

Heat can be covered by the same engine	Months 4–12: April–June heat season with IMD stations and satellite	Heat alerts first, insurance later	Leave heat out; rain only
Competitors (another telecom or a weather firm) move first	Market watch during the pilot	Compete on accuracy in our zones and on Reliance integration	Partner or license; a competitor confirms the market

How the system degrades

SITUATION	WHAT STILL RUNS
Full system	Jio links + network health + radar + gauges (+ satellite, new gauges, paid data only if they earn their cost)
Links thin	Radar + gauges + satellite, network health for Jio operations, gauges on Jio sites
Accuracy short	Soft operational alerts and network operations only; no payouts
No value shown	Stop, with evidence, after about 90 days

Stop criteria

- Gate 1: too few usable links and no telemetry access in the candidate cities → stop the Jio-links track; decide whether a radar-and-gauge decision layer is still worth it.
- Gate 2: no mix clears the accuracy gates → no external scaling; Kavach stays in shadow or is dropped.
- Gate 3: no measurable improvement in a Reliance operation → stop before spending on sales.
- Kavach: no insurer or no payer after shadow mode → drop Kavach; keep intelligence.

IF THEY ASK “WHAT IF YOUR MAIN ASSUMPTION IS WRONG?”

“Then we find out in the first 30 days, cheaply. If Jio’s links are thin, the layer still runs on radar and gauges for Jio’s own operations. If accuracy falls short, we don’t scale and insurance never launches. Every gate is a place to stop with evidence.”

Number guide, never-say list, cheat sheet

Every number and where it comes from

NUMBER	LABEL	SOURCE
944 vs 73 mm	FACT	IMD rainfall, Santacruz vs Colaba, 26 July 2005
15 heavy days, 5 with less than half	OUR CALCULATION	IMD station reports (NOAA archive), 2020–2024 monsoons
Science 2006 · PNAS 2013	FACT	Messer et al. (doi:10.1126/science.1120034) · Overeem et al. (doi:10.1073/pnas.1217961110)
500 links, 11 days	FACT	pycomlink public dataset (KIT), github.com/pycomlink/pycomlink
68% (1,120 of 1,639)	OUR CALCULATION	Our code on that dataset, German Weather Service radar as reference
0.95 · 0.88 · 0.78 · 18% low · 18% unconfirmed	OUR CALCULATION	Same run; Q&A only
80% · ±10% · <15%	OUR TARGET	Pilot gates
10 mm in 3 h, ~4 km zone	OUR ASSUMPTION	Event rule; 5 and 15 mm also tested (64%, 54%)
1.1 / 2.2 / 0.7 / 11.9 days a year	OUR CALCULATION	Counted from IMD station reports, 2000–2024
204.5 / 115.6 / 64.5 mm a day	FACT	IMD rain categories
47°C	OUR ASSUMPTION	Rare-heat trigger (45°C ≈ IMD heatwave level, too frequent)
₹45 / ₹52 / ₹29 / ₹208	OUR CALCULATION	Paid days × ₹300 ÷ 0.6 ÷ 12; ₹29 is our minimum
₹300 · 60% · 5-day cap	OUR ASSUMPTION	Prototype; actuary sets
222% / 242% / 469%; Plus 60%	OUR CALCULATION	Stress test, every year 2000–2024
₹12.5 crore pool; ₹1.9–3.1 crore fees	OUR CALCULATION	50,000 riders × ₹208 × 12; 15–25% fee (illustrative, Q&A only)
90 days, three gates	PLAN	Our pilot design

DATA LINKS

Rain-sensing test: github.com/pycomlink/pycomlink (folder `pycomlink/io/example_data`; also `pip install pycomlink`). Radar reference inside it: German Weather Service RADOLAN YW (opendata.dwd.de). India: noaa-gsod-pds.s3.amazonaws.com/{year}/{station}.csv, stations 43003099999 Santacruz, 43057099999 Colaba, 42867099999 Nagpur, 42647099999 Ahmedabad. Papers: Messer et al., Science 2006 (doi:10.1126/science.1120034); Overeem et al., PNAS 2013 (doi:10.1073/pnas.1217961110).

Never say → say instead

NEVER SAY	SAY INSTEAD
"We tested Jio's network."	"We ran the published method on public research data from Germany."
"Studies show 68%."	"In our own check, links alone caught 68%."
"Jio can already sense the weather." / "Every tower is a rain gauge."	"Jio's microwave links may carry a usable weather signal."

"80% is the industry standard."	"80% is our go/no-go bar, set above the 68% baseline."
"Reliance is our first customer."	"Reliance is our first proving ground."
"₹300 goes to the rider's JioPay wallet."	"A licensed insurer would pay, through its own rail."
"We make ₹174 crore."	Don't raise it. If asked: an illustrative premium pool, not Jio revenue.
"Only Reliance can do this."	"The edge is integration, not the sensor."
"Low cost" / "free" about any source	"The pilot prices each source."
"We built a model."	"We ran the published method and calculated the result ourselves."
"False triggers."	"Triggers not confirmed by radar."
Naming a city for the live test	"The coverage audit selects the city."
"Every phone is a thermometer."	"Heat comes from IMD stations and satellite."
"The network measures temperature near you."	"Links sense rain only."
"₹29 a month for everyone."	"Prices depend on the city; ₹29 is our minimum."
"Zero new hardware."	"Mostly existing infrastructure; gauges only where links are thin."
"Heat will cost India 34 million jobs."	"Working hours equal to about 34 million full-time jobs (ILO)."

The night-before cheat sheet

- 1 India doesn't have a weather-data problem; it has a weather-decision problem.
- 2 Jio's microwave links may carry a weather signal. Fibre doesn't.
- 3 The physics is well established. We tested it ourselves on public German data, not Jio's.
- 4 Links alone caught 68% of rain events (1,120 of 1,639). Our bar is 80%.
- 5 So accuracy decides the mix: links, network health, radar, gauges, satellite. Cheapest mix that clears the bar.
- 6 Reliance is the first proving ground. Hero test: rain or equipment fault, before sending a crew?
- 7 Then sell the same intelligence. Jio sends the alert; the customer decides.
- 8 The edge is integration, not the sensor.
- 9 Insurance is one application: only after the gates, a licensed insurer carries the risk.
- 10 Prices are illustrative: paid days $\times$ ₹300 $\div$ 60% $\div$ 12. ₹208 is not from the 3-hour rule.
- 11 90 days, three gates: coverage, accuracy, value. The audit picks the city. If yes, scale; if not, stop.
- 12 First prove the signal. Then sell the intelligence. Protection comes last.